2026

# Robust Data Modeling with Ensembling and Tuning Techniques

PROJECT: EASYVISA: VISA APPLICATION CERTIFICATION PREDICTION

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GREAT LEARNINGS

## Table of Contents


1. Executive Summary
2. Business Problem and Objective
3. Dataset Overview
4. Exploratory Data Analysis (EDA)
    - 4.1 Univariate Analysis
    - 4.2 Bivariate Analysis
    - 4.3 Correlation Analysis
5. Data Preprocessing
6. Model Building and Evaluation
    - 6.1 Original Data Models
    - 6.2 Oversampled Data Models
    - 6.3 Undersampled Data Models
7. Hyperparameter Tuning
8. Final Model Selection

9.Actionable Insights and Recommendations

10. Limitations

11. Conclusion

## 1. Executive Summary

The objective of this project is to develop a Machine Learning classification model capable of predicting whether a visa application is likely to be **Certified** or **Denied**.

The rapid increase in visa applications has made manual screening increasingly difficult and time-consuming. To improve operational efficiency, EasyVisa aims to leverage historical visa application data and predictive analytics to support faster and more accurate visa certification decisions.

The dataset contains applicant-related, employer-related, and job-related information such as education level, work experience, prevailing wage, region of employment, and company details.

Exploratory Data Analysis (EDA) revealed that variables such as education level, previous work experience, prevailing wage, wage unit, continent, and employment region significantly influence visa certification outcomes.

Multiple Machine Learning models were developed and compared using:

- Original data
- Oversampled data
- Undersampled data

Hyperparameter tuning was performed on selected ensemble models to further improve performance.

The final selected model demonstrated strong predictive capability and can support OFLC and EasyVisa in prioritizing applications, reducing processing delays, and improving consistency in visa screening decisions.

## 2. Business Problem and Objective

Business organizations in the United States frequently depend on skilled foreign workers to address workforce shortages and maintain competitiveness. The Office of Foreign Labor Certification (OFLC) processes a large number of visa applications each year, making manual evaluation increasingly resource-intensive.

The increasing number of visa applications creates the need for a Machine Learning-based solution that can assist in:

- Predicting visa approval likelihood
- Identifying applications requiring deeper review
- Reducing manual screening effort
- Improving operational efficiency
- Supporting workforce planning decisions

This project focuses on building predictive classification models using historical visa application data.

The target variable in this project is:

- case_status
    - Certified
    - Denied

| case_status | count |
|---|---|
| Certified | 17018 |
| Denied | 8462 |

| case_status | proportion |
|---|---|
| Certified | 66.789639 |
| Denied | 33.210361 |

The project aims to provide a reliable decision-support system while maintaining human oversight in the final approval process.

## 3. Dataset Overview

The dataset contains **25,480 visa application records** and multiple applicant-related, employer-related, and job-related variables.

The dataset includes the following features:

| Variable | Description |
|---|---|
| case_id | Unique application identifier |
| continent | Applicant continent |
| education_of_employee | Education level of applicant |
| has_job_experience | Previous work experience status |
| requires_job_training | Job training requirement |
| no_of_employees | Number of employees in employer company |
| yr_of_estab | Employer company establishment year |
| region_of_employment | Intended employment region |
| prevailing_wage | Wage offered |
| unit_of_wage | Wage unit |
| full_time_position | Full-time or part-time position |
| case_status | Target variable |

The dataset showed no missing values. However:

- Invalid negative values were identified in no_of_employees
- Numerical variables contained significant outliers

- The target variable displayed moderate class imbalance

## Dataset Shape and Information

```
(25480, 12)
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25480 entries, 0 to 25479
Data columns (total 12 columns):
 #   Column                 Non-Null Count  Dtype  
---  ------                 --------------  -----  
 0   case_id                25480 non-null  object 
 1   continent              25480 non-null  object 
 2   education_of_employee  25480 non-null  object 
 3   has_job_experience     25480 non-null  object 
 4   requires_job_training  25480 non-null  object 
 5   no_of_employees        25480 non-null  int64  
 6   yr_of_estab            25480 non-null  int64  
 7   region_of_employment   25480 non-null  object 
 8   prevailing_wage        25480 non-null  float64
 9   unit_of_wage           25480 non-null  object 
 10  full_time_position     25480 non-null  object 
 11  case_status            25480 non-null  object 
dtypes: float64(1), int64(2), object(9)
memory usage: 2.3+ MB
```

## Missing Values Summary

| | 0 |
|---|---|
| case_id | 0 |
| continent | 0 |
| education_of_employee | 0 |
| has_job_experience | 0 |
| requires_job_training | 0 |
| no_of_employees | 0 |
| yr_of_estab | 0 |
| region_of_employment | 0 |
| prevailing_wage | 0 |
| unit_of_wage | 0 |
| full_time_position | 0 |
| case_status | 0 |

dtype: int64

## 4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand data distributions, identify important patterns, detect anomalies, and examine relationships between variables and visa certification outcomes.

### 4.1 Univariate Analysis

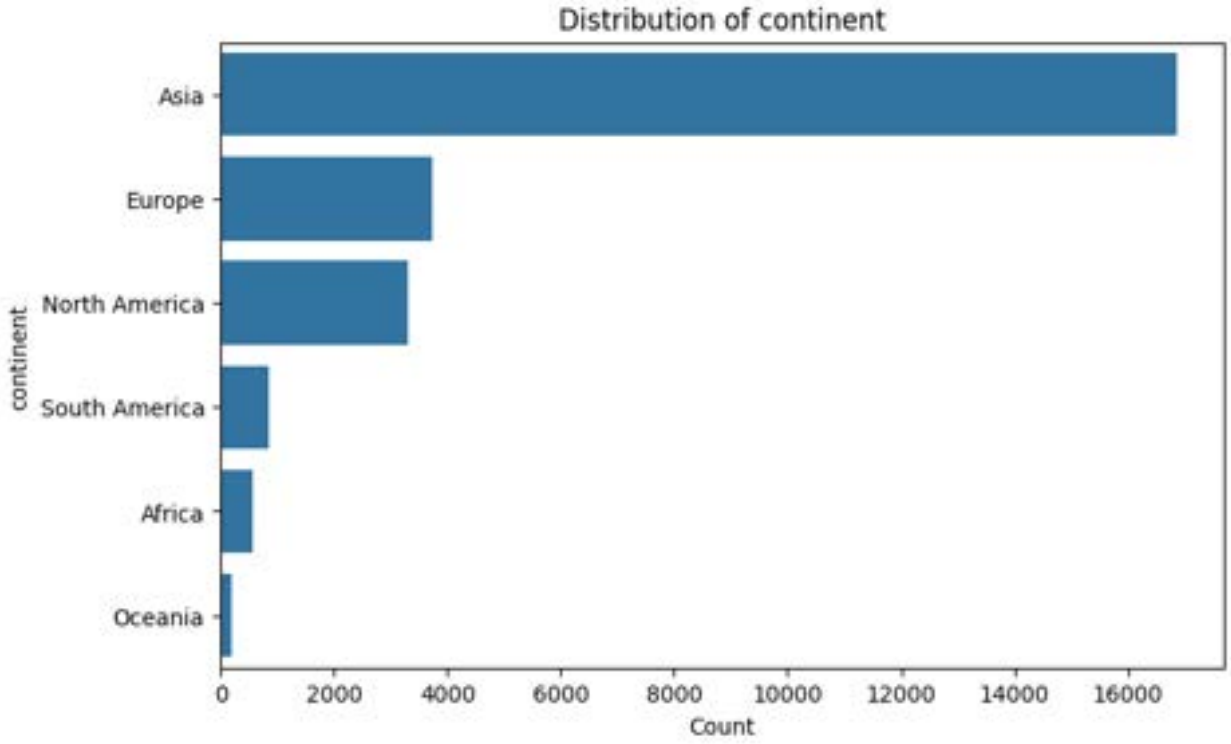
Univariate analysis focused on understanding the distribution of individual variables.

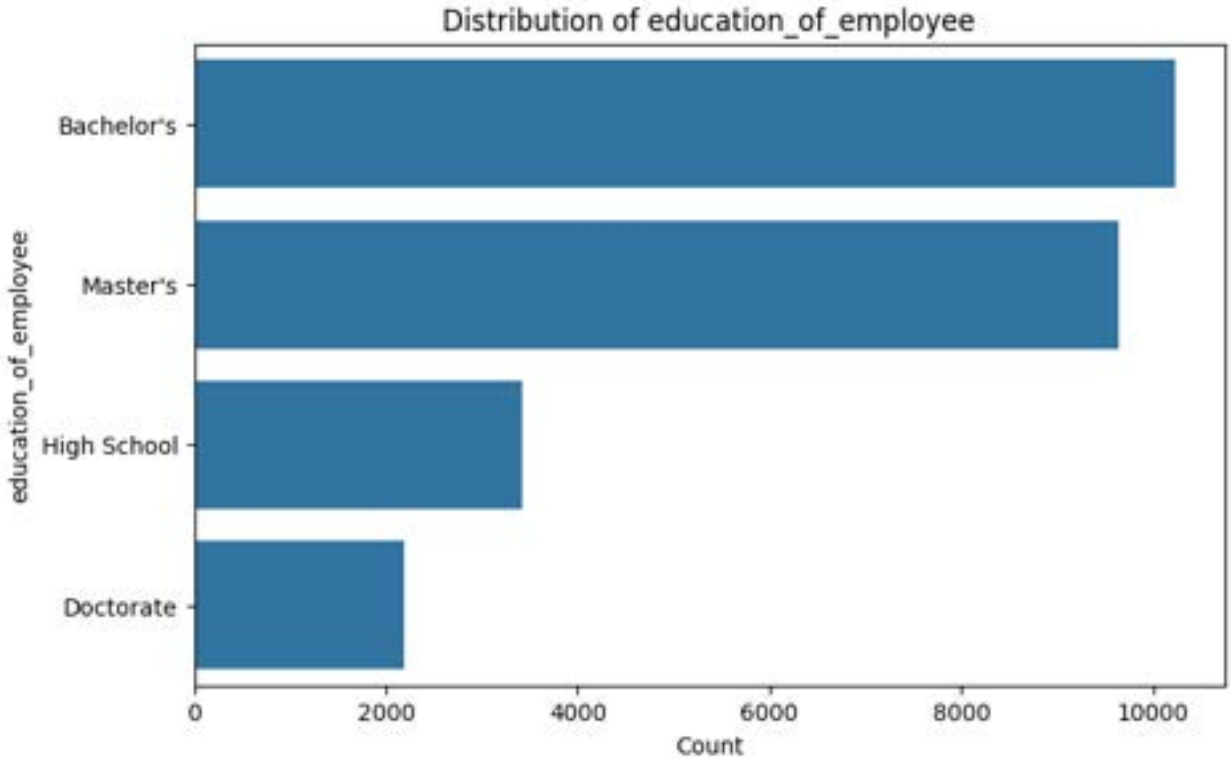
The target variable analysis showed that Certified applications were more common than Denied applications, indicating moderate imbalance.

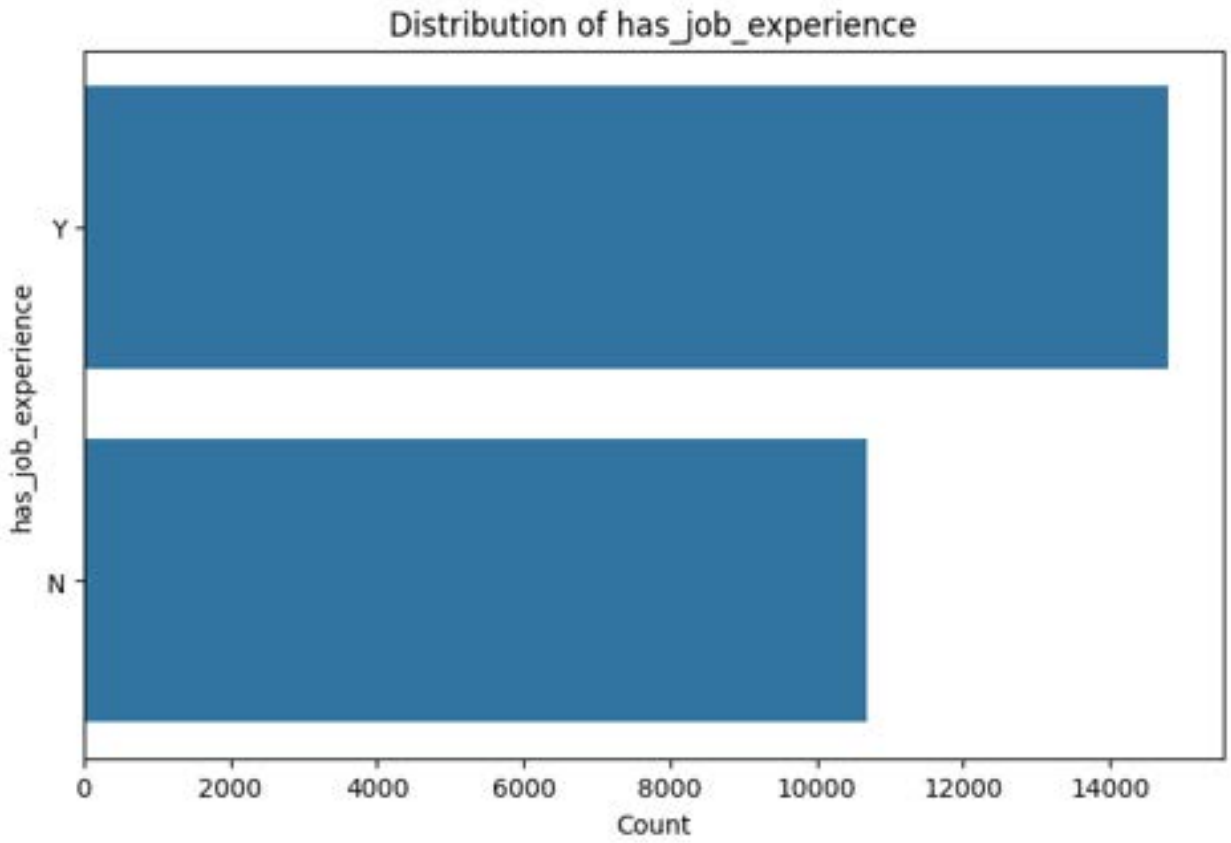
Most applicants held higher education qualifications, and most positions were full-time.

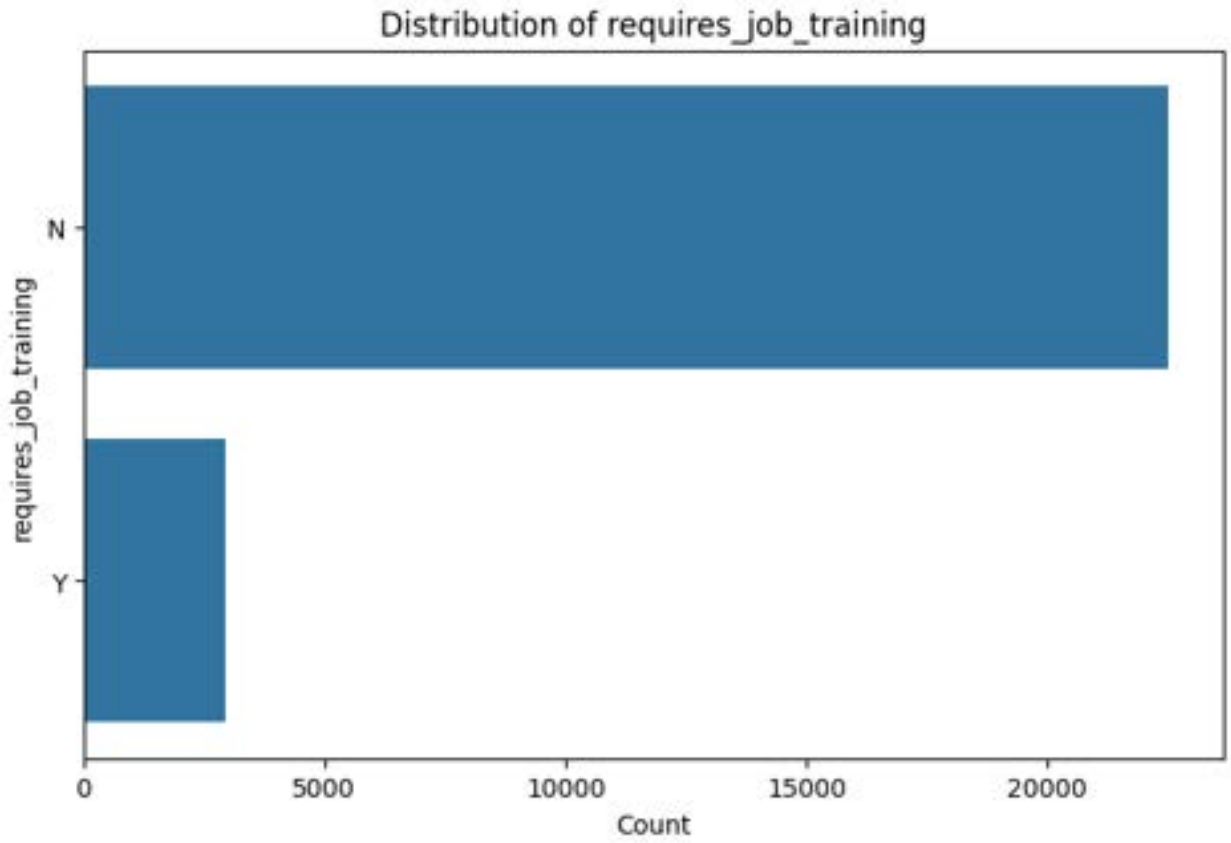
The prevailing wage variable displayed a highly skewed distribution with extreme outliers.

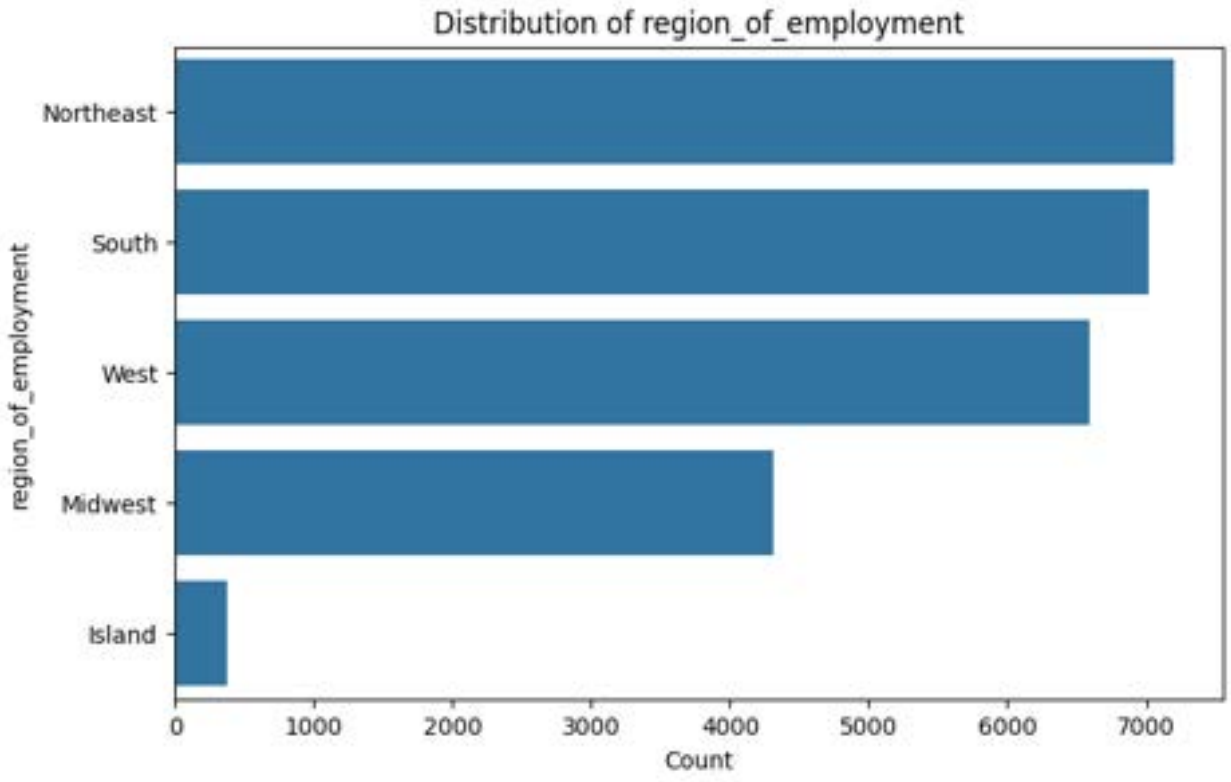
Hourly wage applications represented a relatively smaller portion of the dataset compared to yearly wage applications.

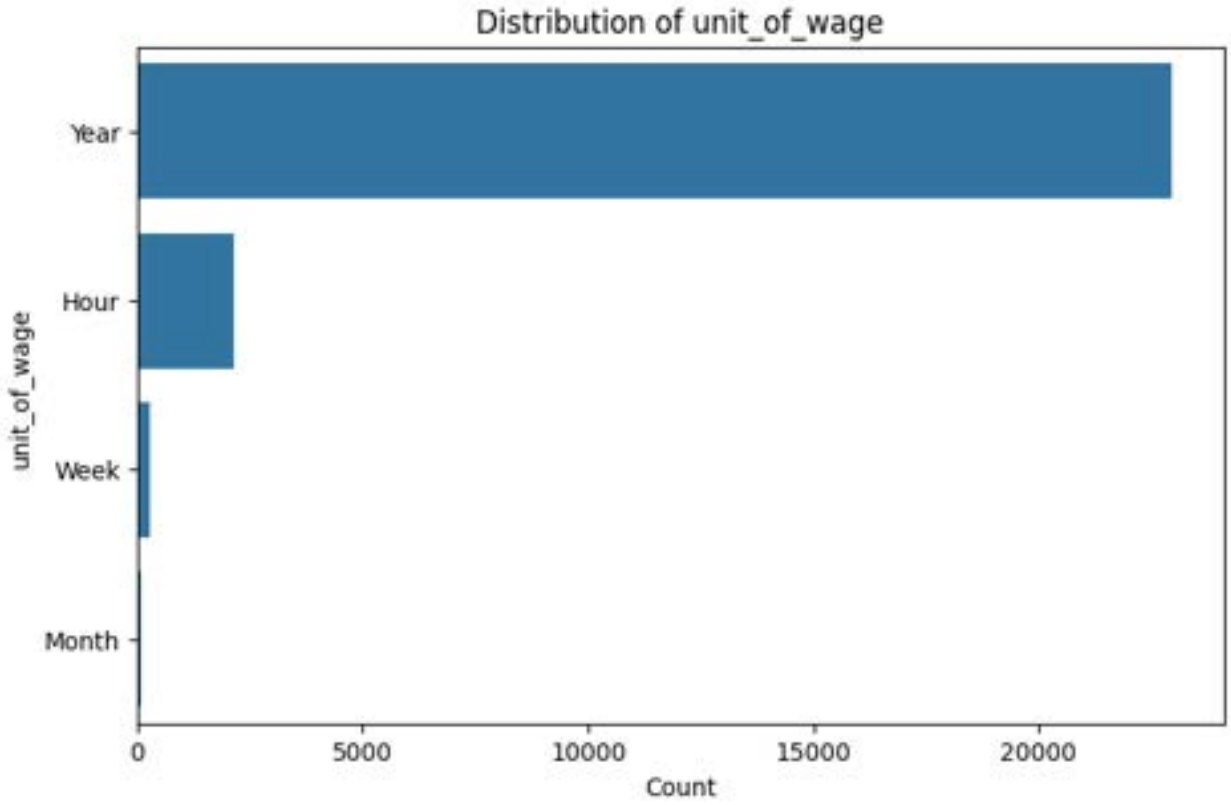

Distribution of continent
continent
Asia
Europe
North America
South America
Africa
Oceania
0
2000
4000
6000
8000
10000
12000
14000
16000
Count

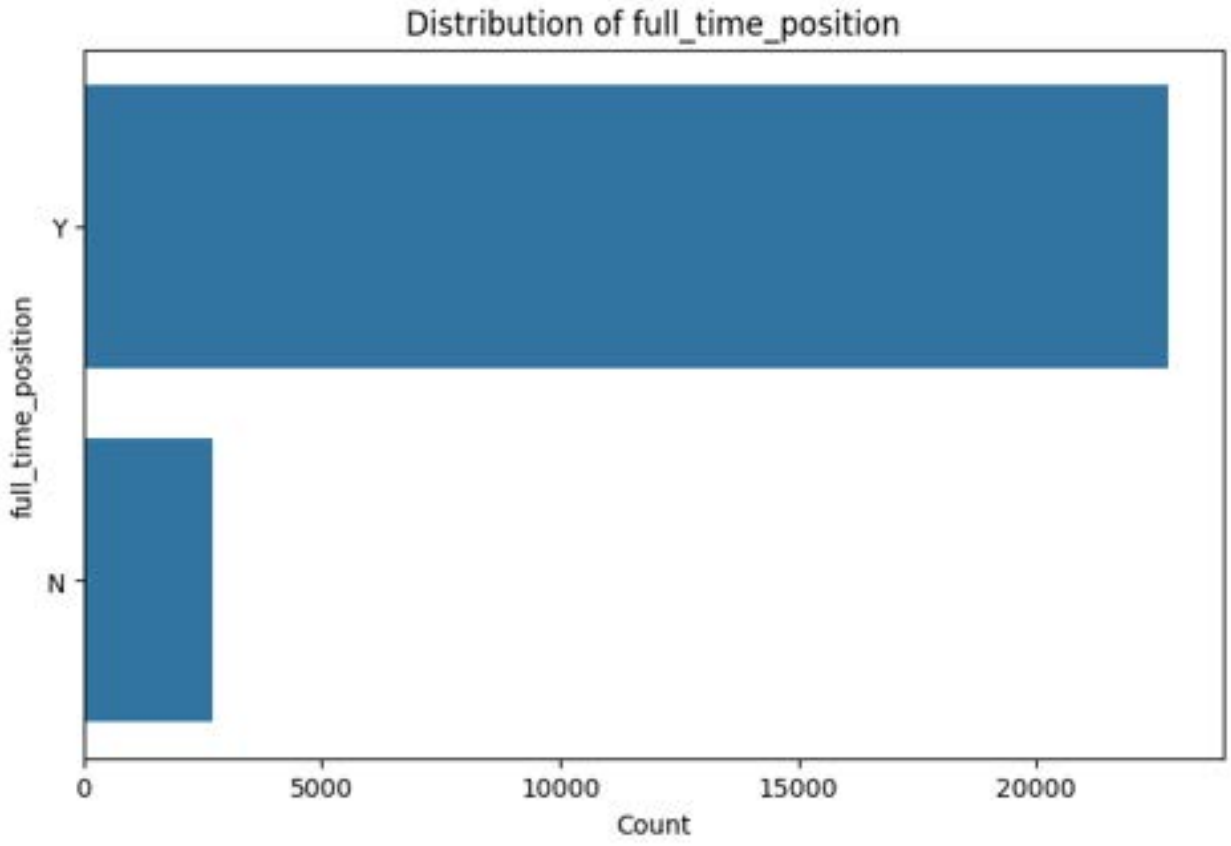

Distribution of education_of_employee
education_of_employee
Bachelor's
Master's
High School
Doctorate
0
2000
4000
6000
8000
10000
Count

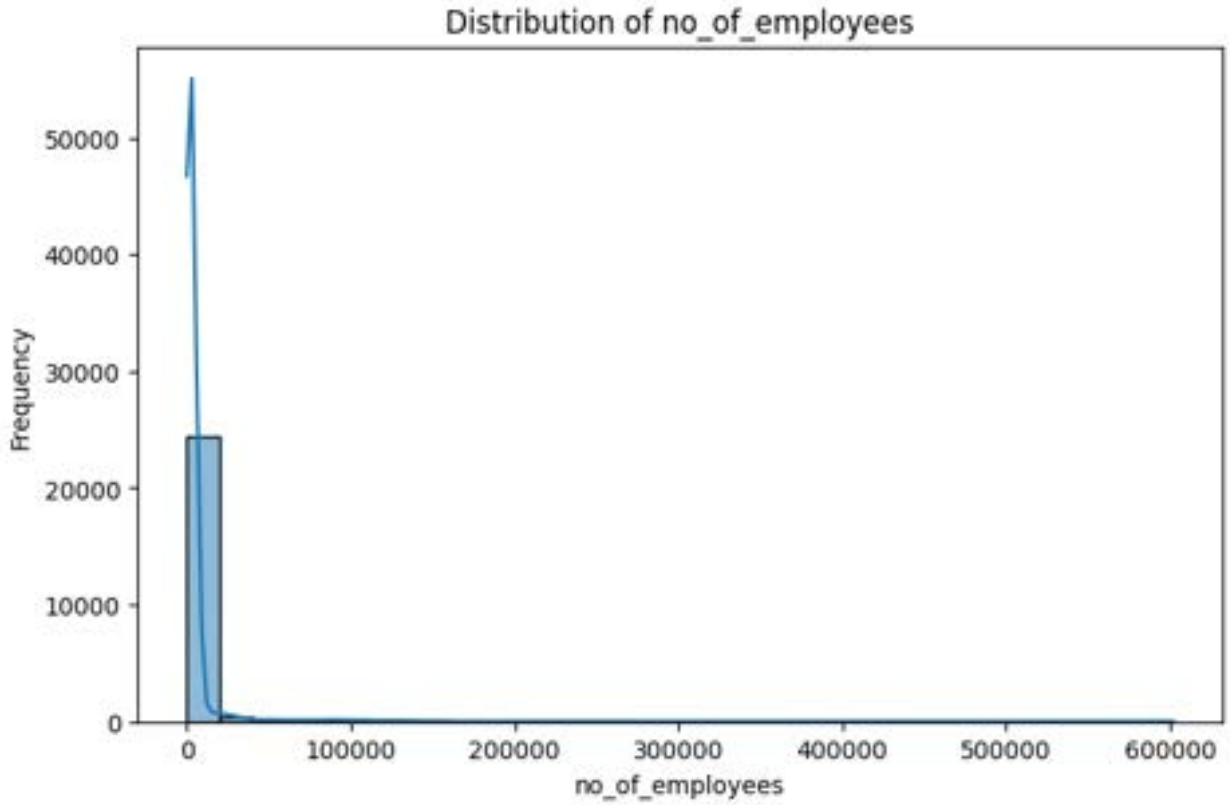

Distribution of has_job_experience
has_job_experience
Y
N
0
2000
4000
6000
8000
10000
12000
14000
Count

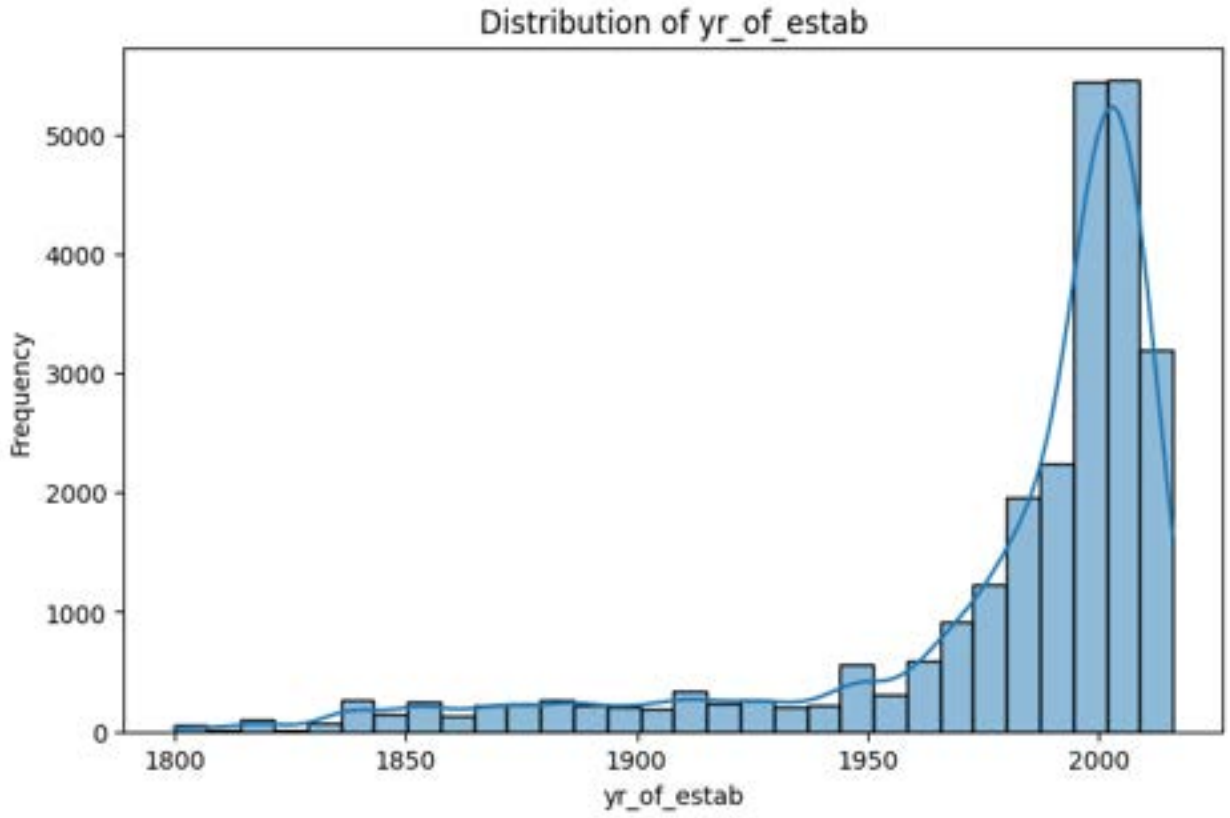

Distribution of requires_job_training
requires_job_training
N
Y
0
5000
10000
15000
20000
Count

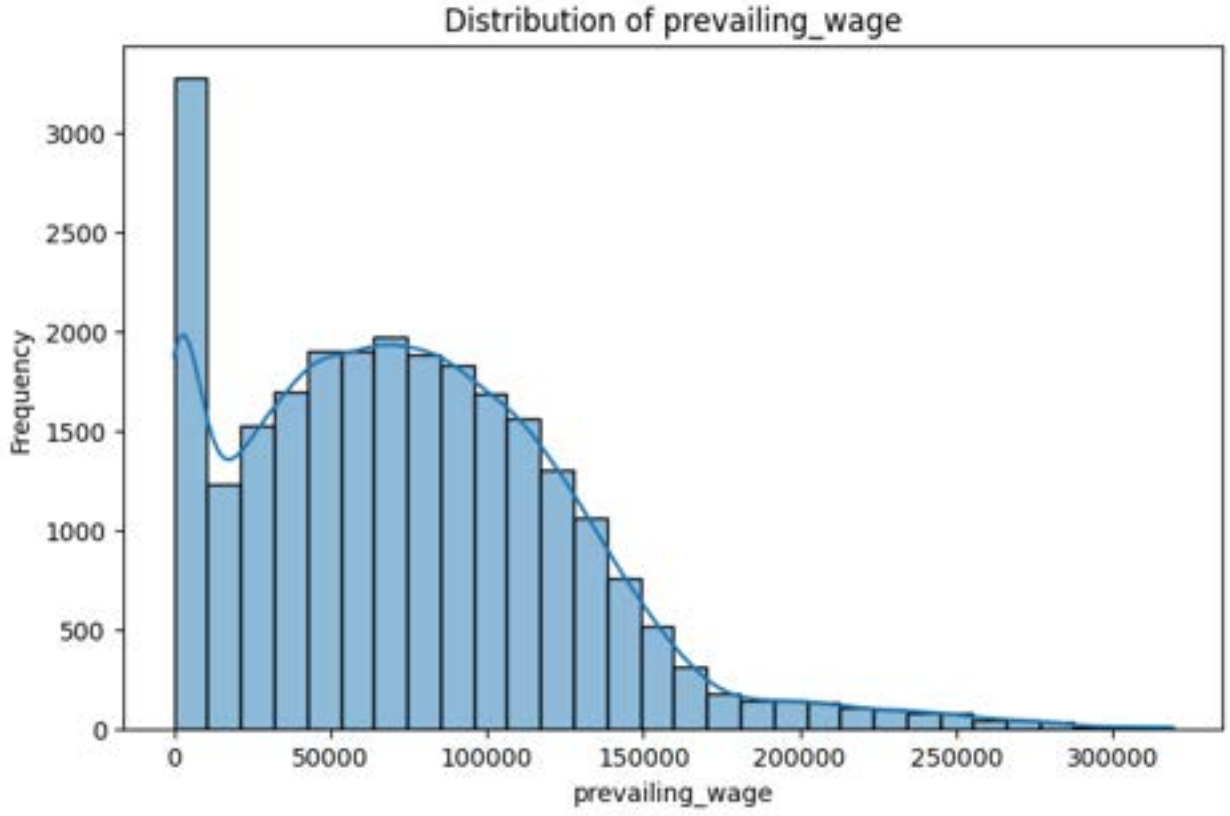

Distribution of region_of_employment
region_of_employment
Northeast
South
West
Midwest
Island
0
1000
2000
3000
4000
5000
6000
7000
Count


Distribution of unit_of_wage
Year
Hour
Week
Month
unit_of_wage
0
5000
10000
15000
20000
Count


Distribution of full_time_position
Y
N
full_time_position
0
5000
10000
15000
20000
Count


Distribution of no_of_employees
Frequency
50000
40000
30000
20000
10000
0
0
100000
200000
300000
400000
500000
600000
no_of_employees


Distribution of yr_of_estab
Frequency
0
1000
2000
3000
4000
5000
1800
1850
1900
1950
2000
yr_of_estab


Distribution of prevailing_wage
Frequency
0
500
1000
1500
2000
2500
3000
0
50000
100000
150000
200000
250000
300000
prevailing_wage

Univariate Observations

- The majority of examples come from the Asian region.
- People with bachelor's or master's degrees account for a major share of applicants.
- Most applicants have employment opportunities.
- Examples typically contain annual salary information.
- Variable prevailing_wage exhibits a positive skewness with some high-value outliers.
- Variable no_of_employees is extremely skewed, containing some extreme values.
- Variable no_of_employees includes negative values, which cannot logically occur and need to be fixed at the data preprocessing stage.
- Variable case_id is just an identifier that shouldn't be used as an explanatory variable.

## 4.2 Bivariate Analysis

Bivariate analysis was performed to evaluate relationships between explanatory variables and visa certification outcomes.

Applicants with previous work experience demonstrated higher certification rates compared to applicants without experience.

Applicants with higher education levels also showed stronger certification outcomes.

Applications with yearly wage units generally had higher approval rates, whereas hourly wage applications displayed relatively higher denial rates.

Certification rates varied across continents and regions of employment.

Certified applications generally exhibited higher prevailing wages than denied applications.

Certification Rate by continent

| case_status | Certified | Denied | Total_Count |
|---|---|---|---|
| **continent** | | | |
| **Europe** | 79.233655 | 20.766345 | 3732 |
| **Africa** | 72.050817 | 27.949183 | 551 |
| **Asia** | 65.310480 | 34.689520 | 16861 |
| **Oceania** | 63.541667 | 36.458333 | 192 |
| **North America** | 61.877278 | 38.122722 | 3292 |
| **South America** | 57.863850 | 42.136150 | 852 |

Certification Rate by education_of_employee

| case_status | Certified | Denied | Total_Count |
|---|---|---|---|
| **education_of_employee** | | | |
| **Doctorate** | 87.226277 | 12.773723 | 2192 |
| **Master's** | 78.627777 | 21.372223 | 9634 |
| **Bachelor's** | 62.214188 | 37.785812 | 10234 |
| **High School** | 34.035088 | 65.964912 | 3420 |

Certification Rate by has_job_experience

| case_status | Certified | Denied | Total_Count |
|---|---|---|---|
| **has_job_experience** | | | |
| **Y** | 74.476422 | 25.523578 | 14802 |
| **N** | 56.134108 | 43.865892 | 10678 |

Certification Rate by requires_job_training

| case_status | Certified | Denied | Total_Count |
|---|---|---|---|
| **requires_job_training** | | | |
| **Y** | 67.884941 | 32.115059 | 2955 |
| **N** | 66.645949 | 33.354051 | 22525 |

Certification Rate by region_of_employment

| case_status | Certified | Denied | Total_Count |
|---|---|---|---|
| **region_of_employment** | | | |
| **Midwest** | 75.528210 | 24.471790 | 4307 |
| **South** | 70.015676 | 29.984324 | 7017 |
| **Northeast** | 62.904795 | 37.095205 | 7195 |

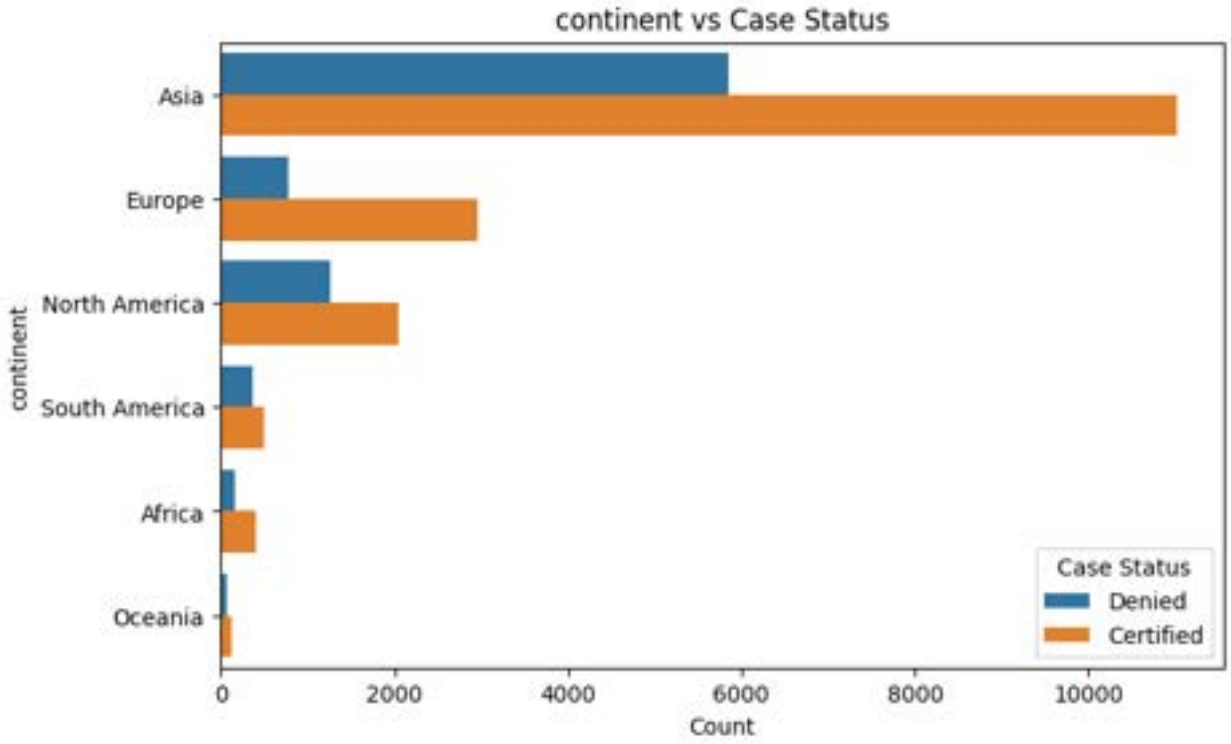
| case_status | Certified | Denied | Total_Count |
|---|---|---|---|
| **region_of_employment** | | | |
| **West** | 62.253265 | 37.746735 | 6586 |
| **Island** | 60.266667 | 39.733333 | 375 |

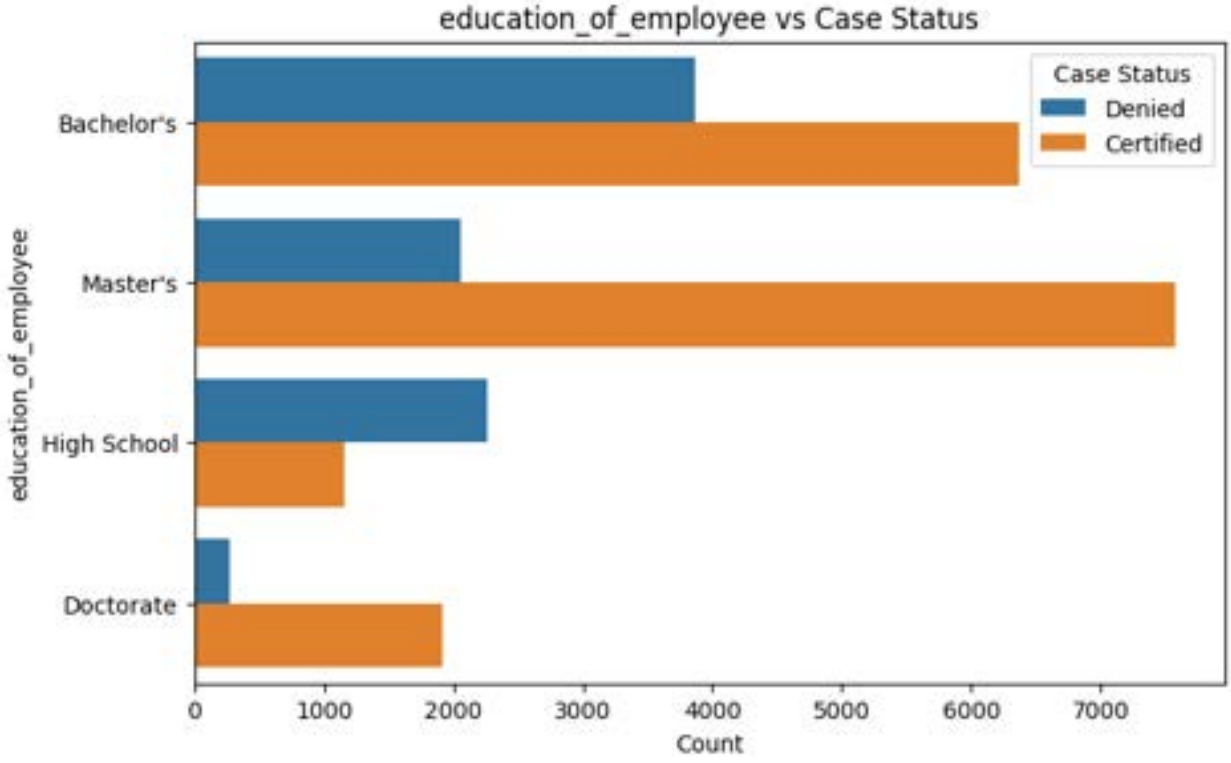
Certification Rate by unit_of_wage

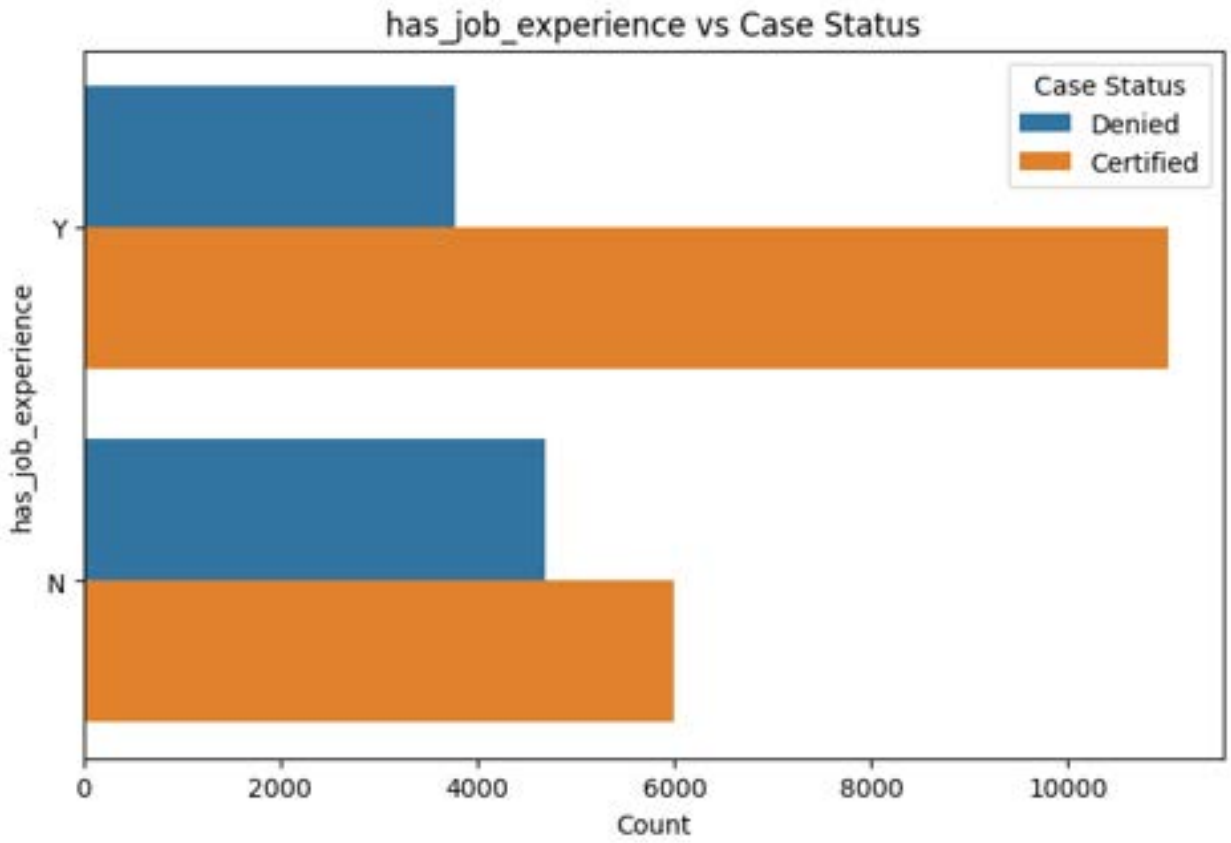
| case_status | Certified | Denied | Total_Count |
|---|---|---|---|
| **unit_of_wage** | | | |
| **Year** | 69.885027 | 30.114973 | 22962 |
| **Week** | 62.132353 | 37.867647 | 272 |
| **Month** | 61.797753 | 38.202247 | 89 |
| **Hour** | 34.631433 | 65.368567 | 2157 |

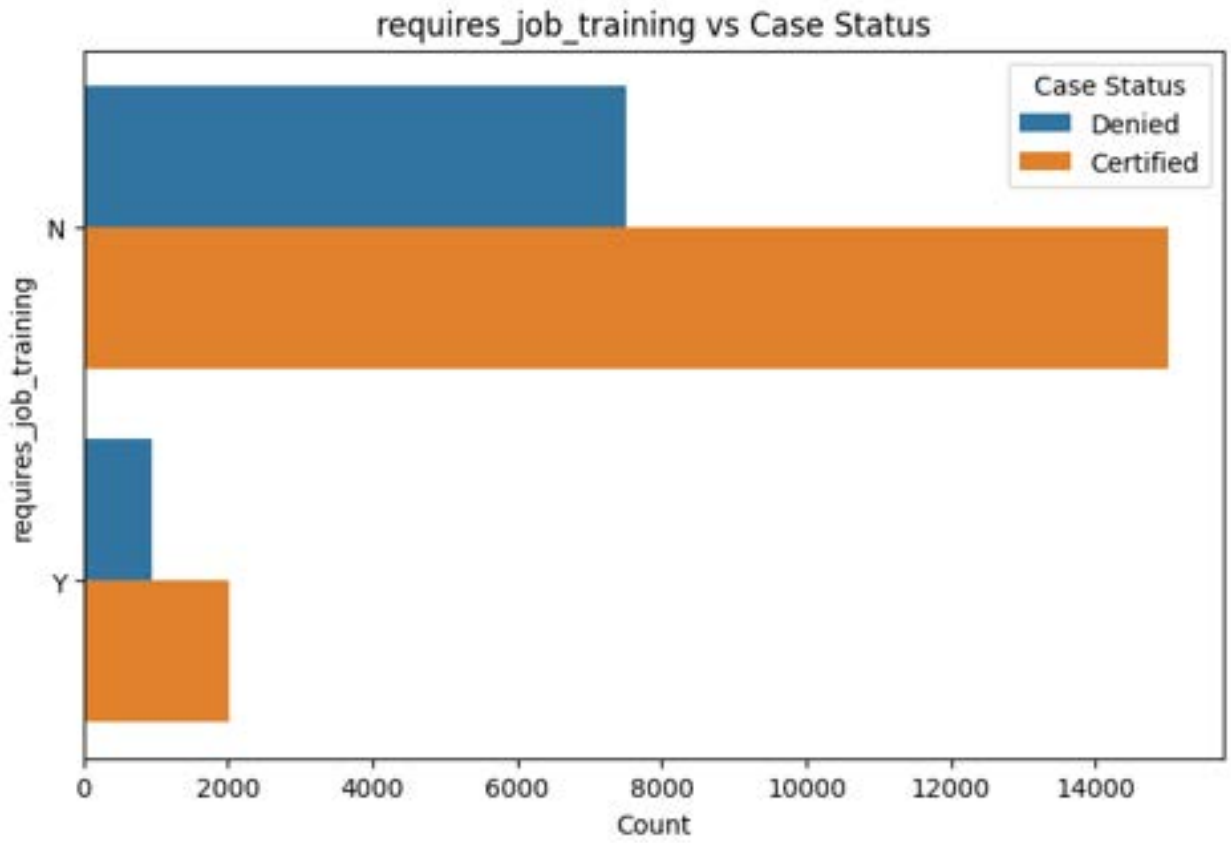
Certification Rate by full_time_position

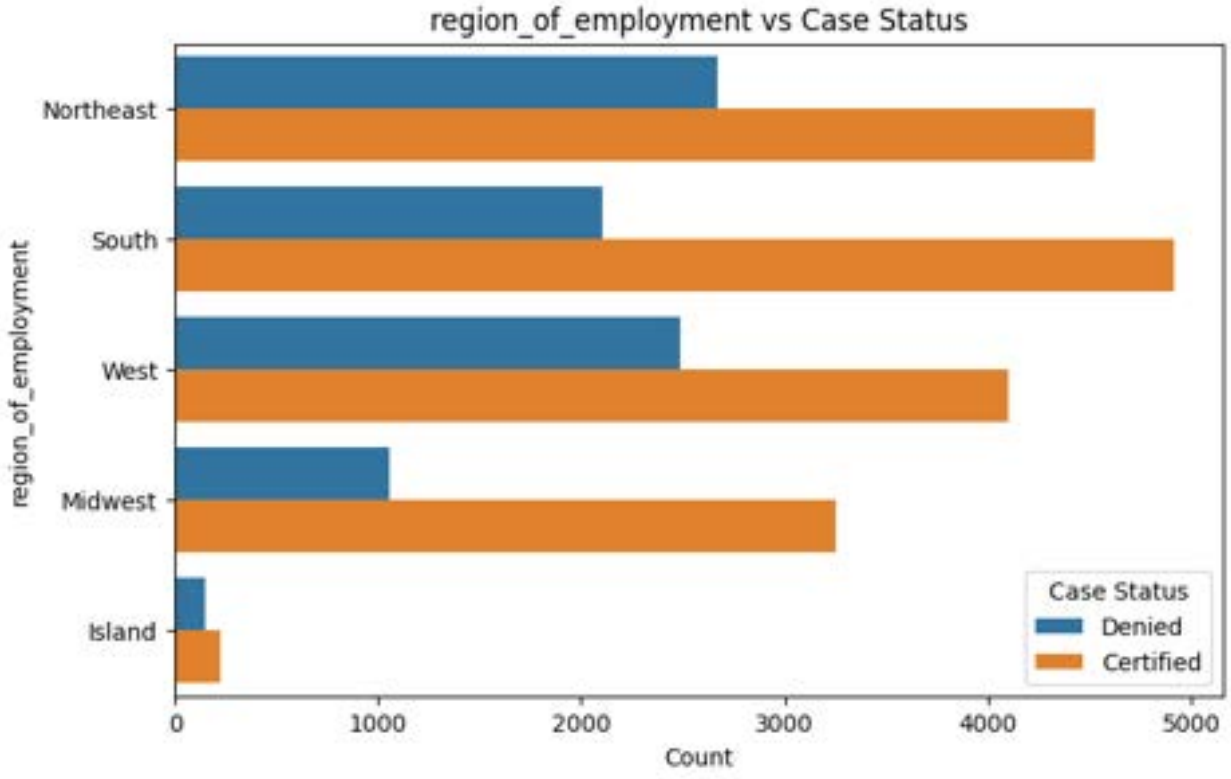
| case_status | Certified | Denied | Total_Count |
|---|---|---|---|
| **full_time_position** | | | |
| **N** | 68.526044 | 31.473956 | 2707 |
| **Y** | 66.583235 | 33.416765 | 22773 |

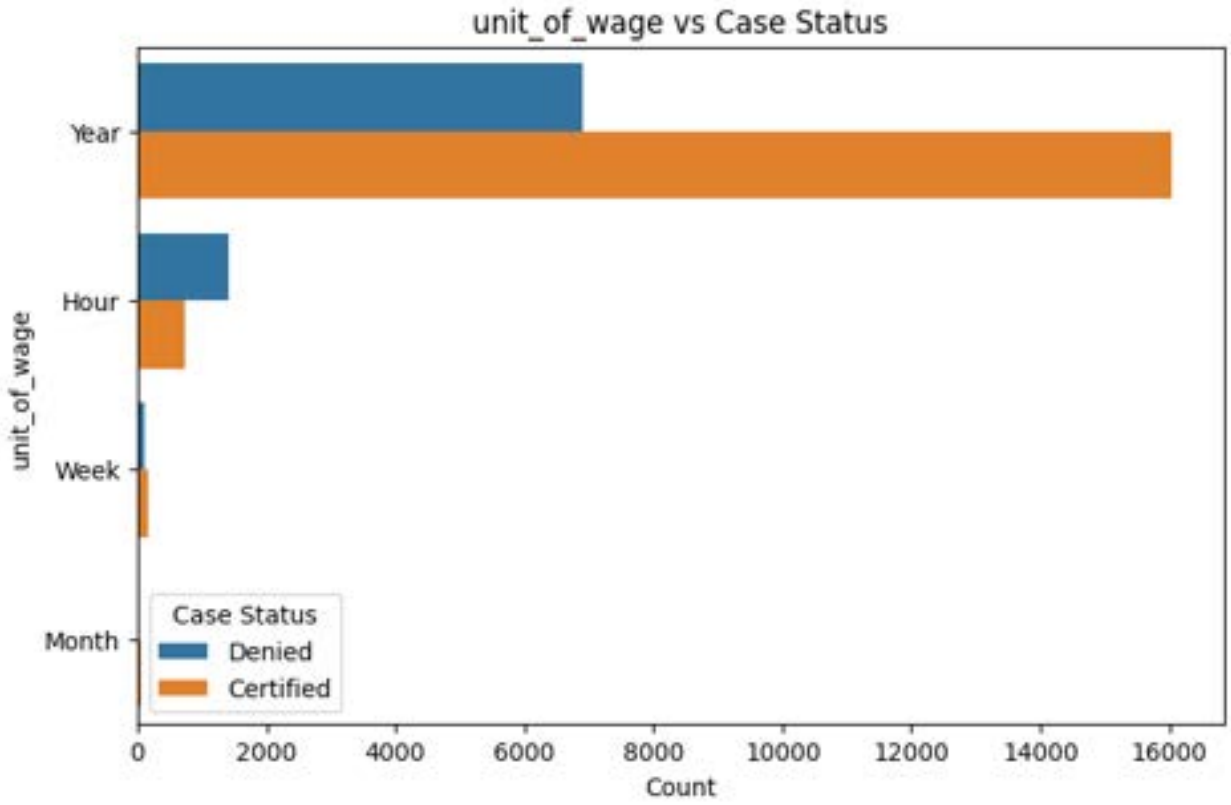

continent vs Case Status
Asia
Europe
North America
South America
Africa
Oceania
continent
Case Status
Denied
Certified
0
2000
4000
6000
8000
10000
Count

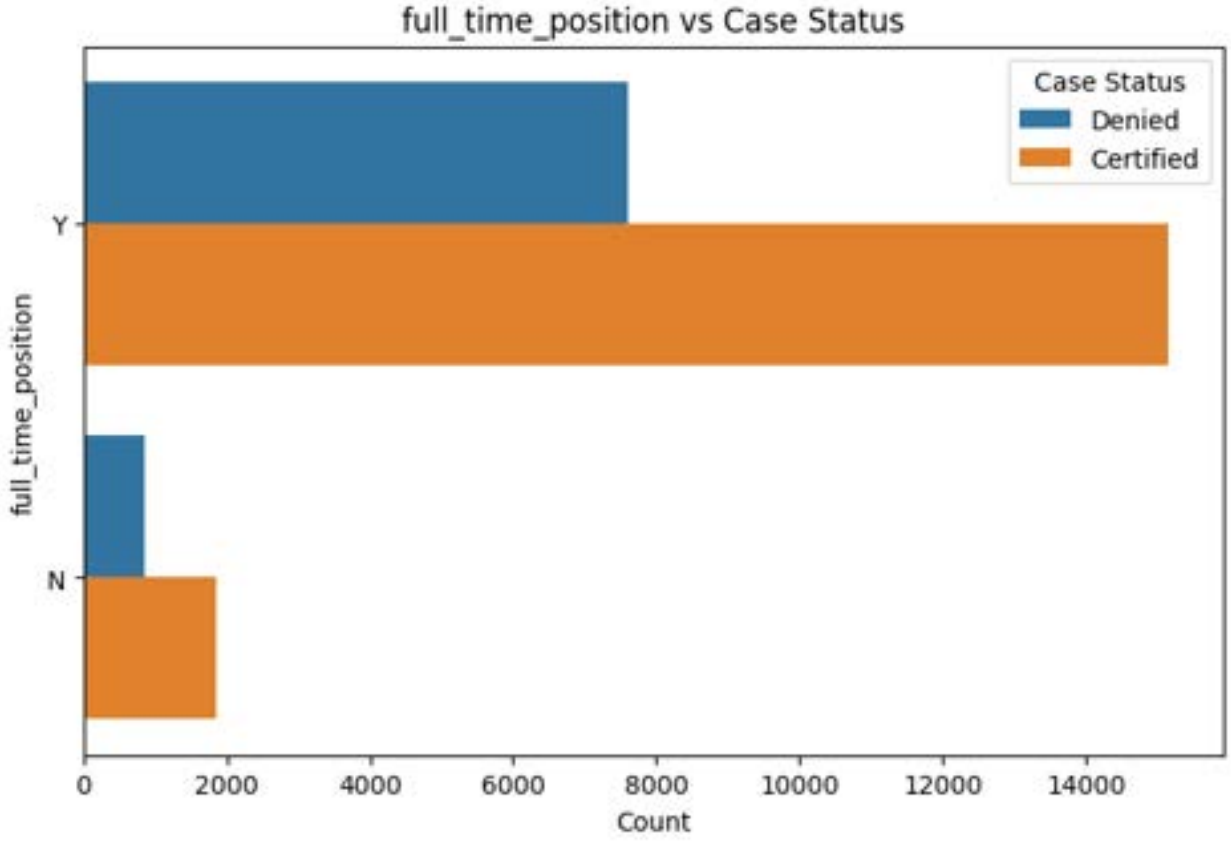

education_of_employee vs Case Status
Bachelor's
Master's
High School
Doctorate
education_of_employee
Case Status
Denied
Certified
0
1000
2000
3000
4000
5000
6000
7000
Count

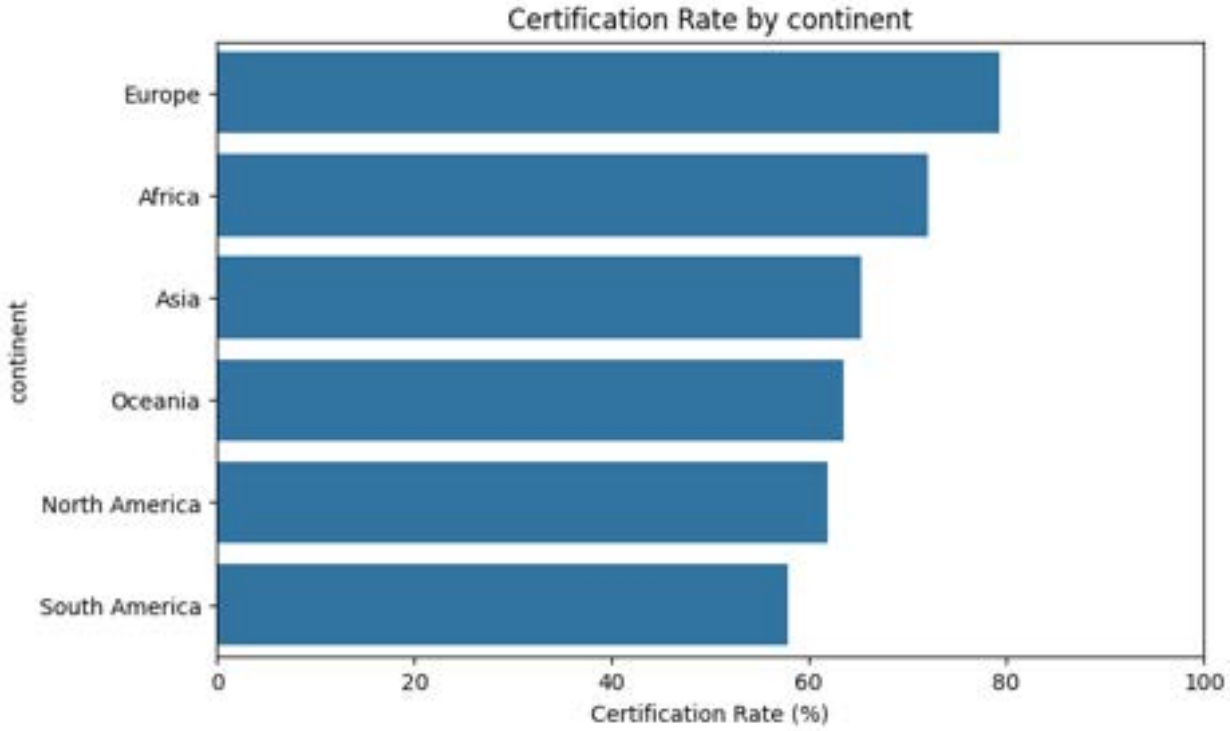

has_job_experience vs Case Status
Case Status
Denied
Certified
has_job_experience
Y
N
0
2000
4000
6000
8000
10000
Count

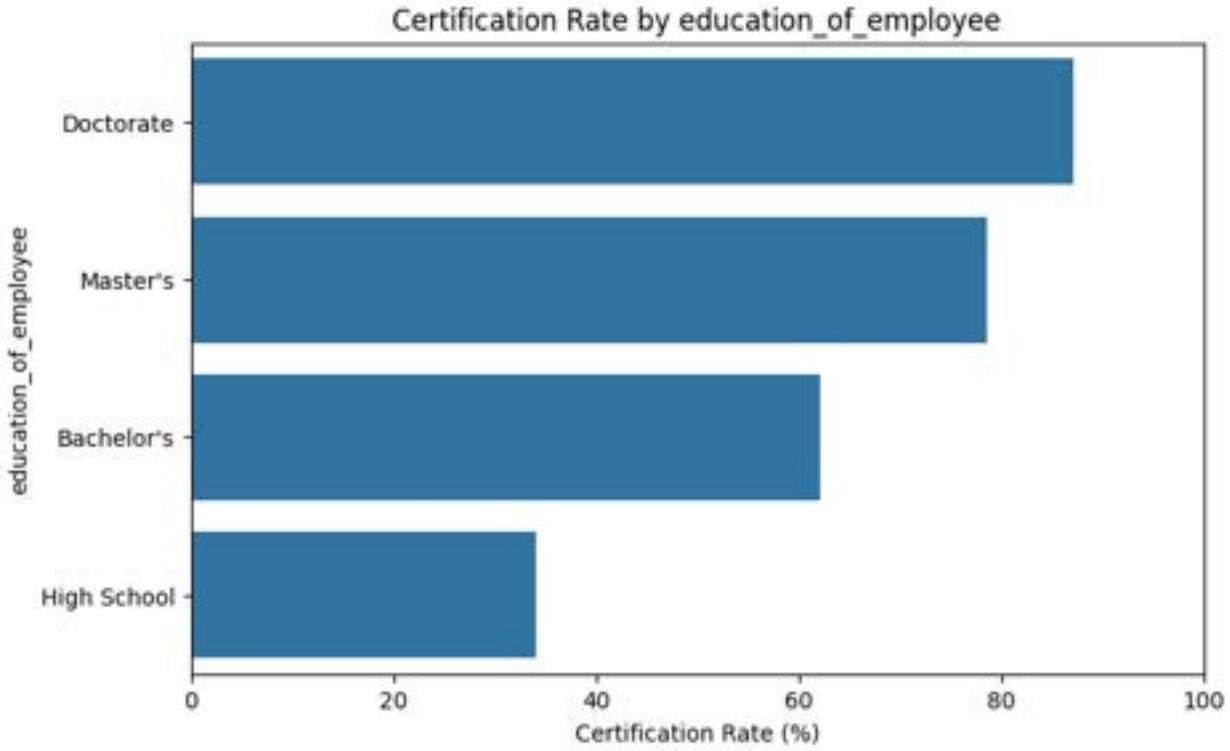

requires_job_training vs Case Status
Case Status
Denied
Certified
requires_job_training
N
Y
0
2000
4000
6000
8000
10000
12000
14000
Count

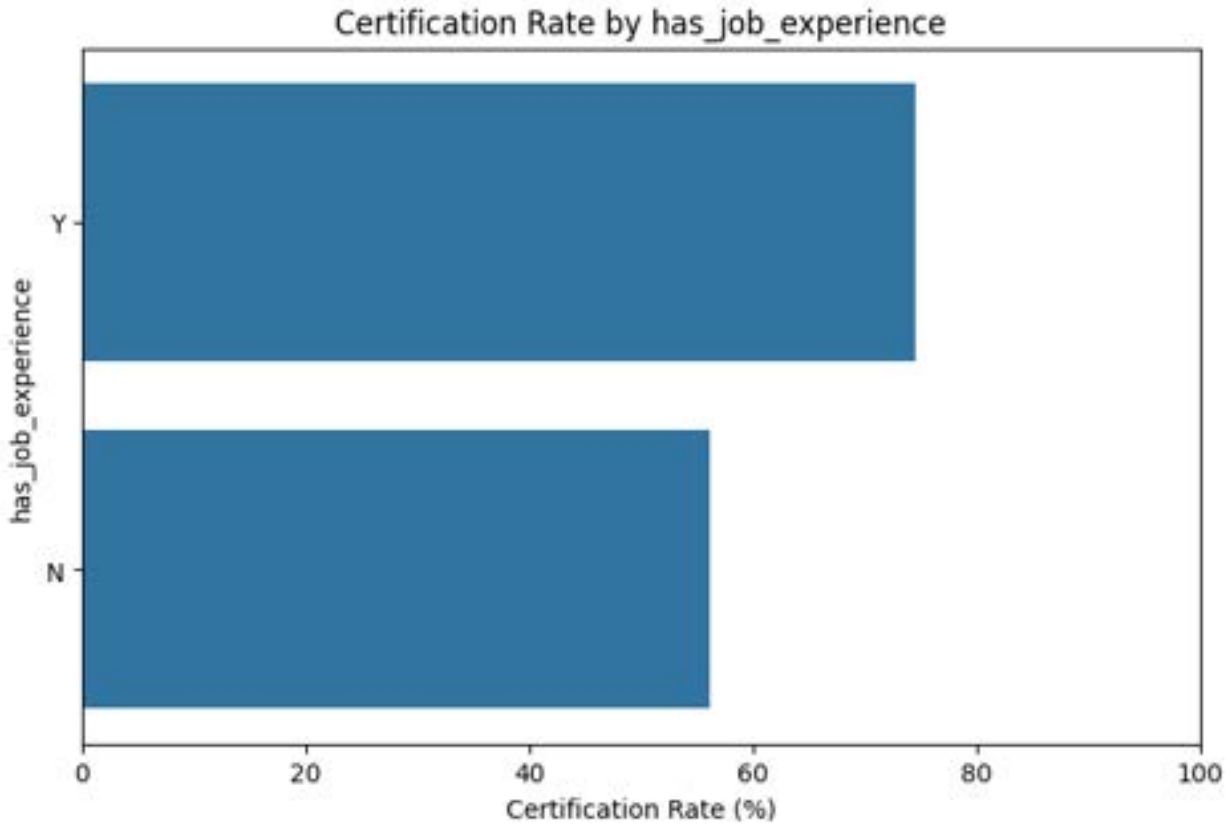

region_of_employment vs Case Status
region_of_employment
Northeast
South
West
Midwest
Island
Case Status
Denied
Certified
0
1000
2000
3000
4000
5000
Count

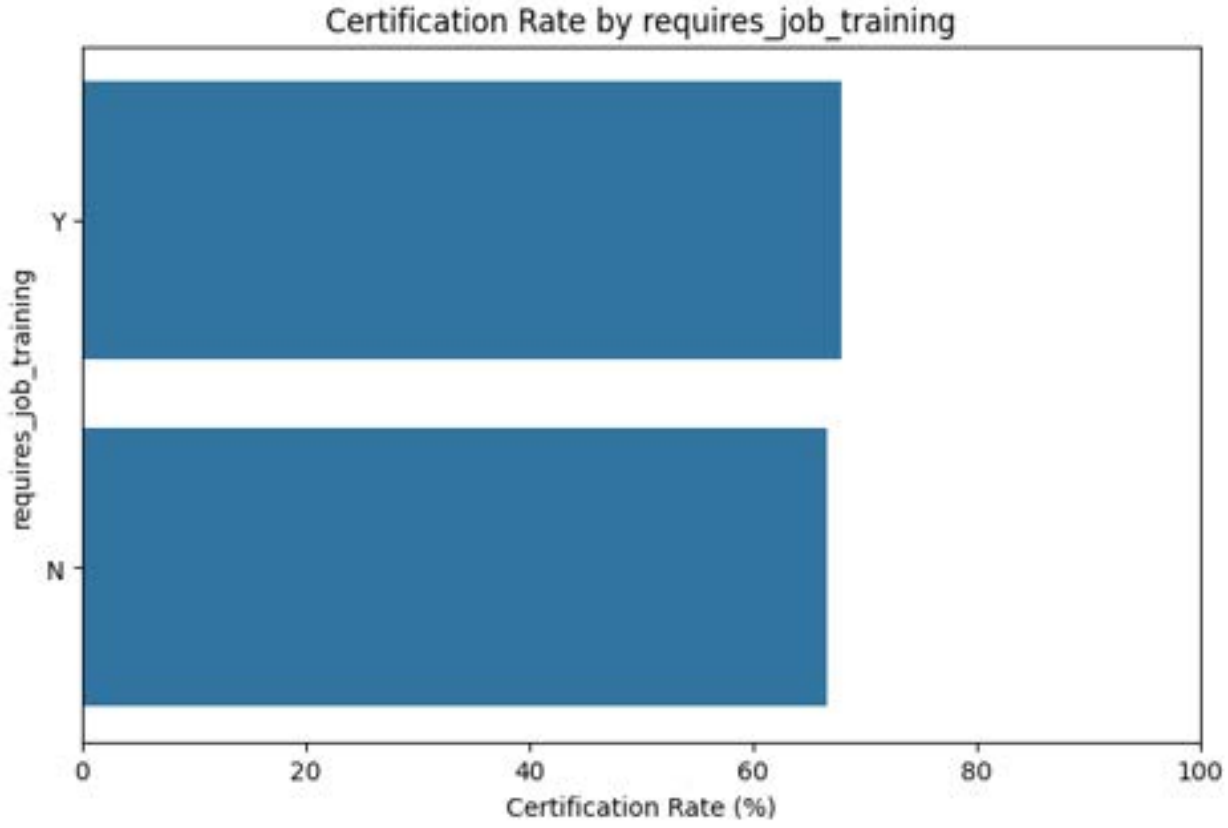

unit_of_wage vs Case Status
Year
Hour
Week
Month
unit_of_wage
Case Status
Denied
Certified
0
2000
4000
6000
8000
10000
12000
14000
16000
Count

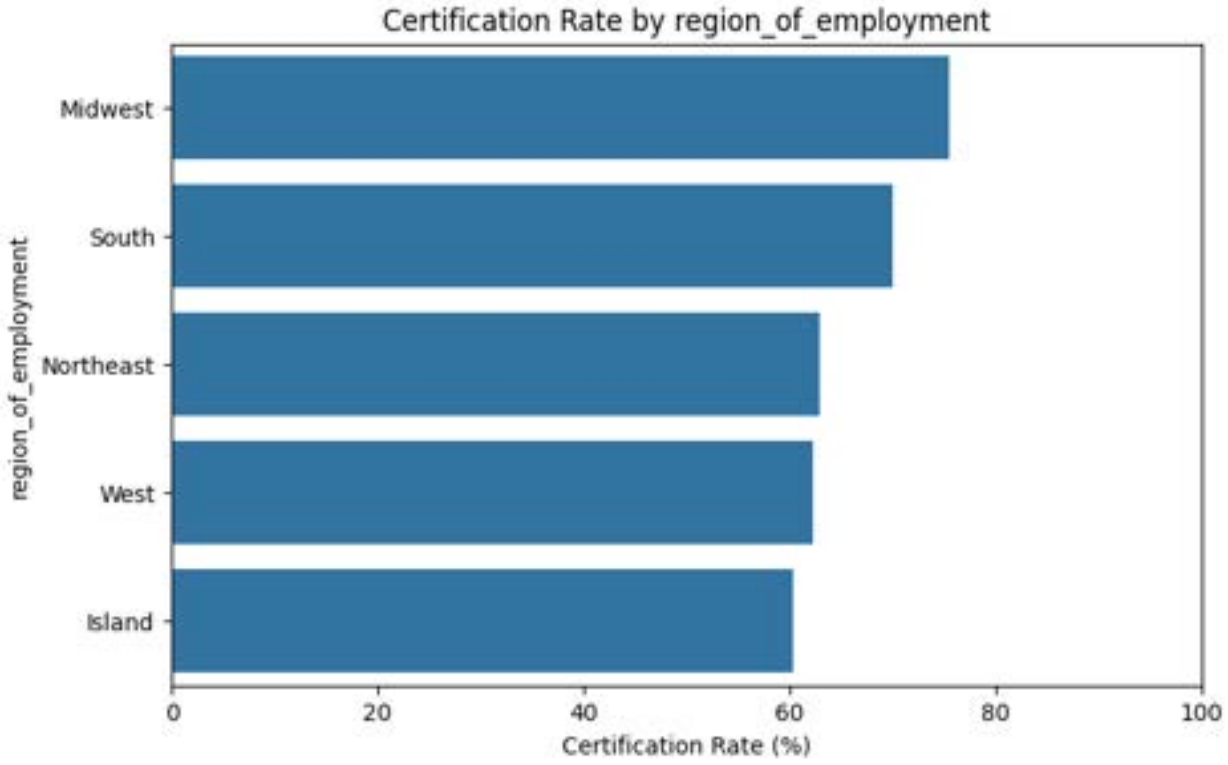

full_time_position vs Case Status
Y
N
full_time_position
Case Status
Denied
Certified
0
2000
4000
6000
8000
10000
12000
14000
Count


Certification Rate by continent
Europe
Africa
Asia
Oceania
North America
South America
continent
0
20
40
60
80
100
Certification Rate (%)

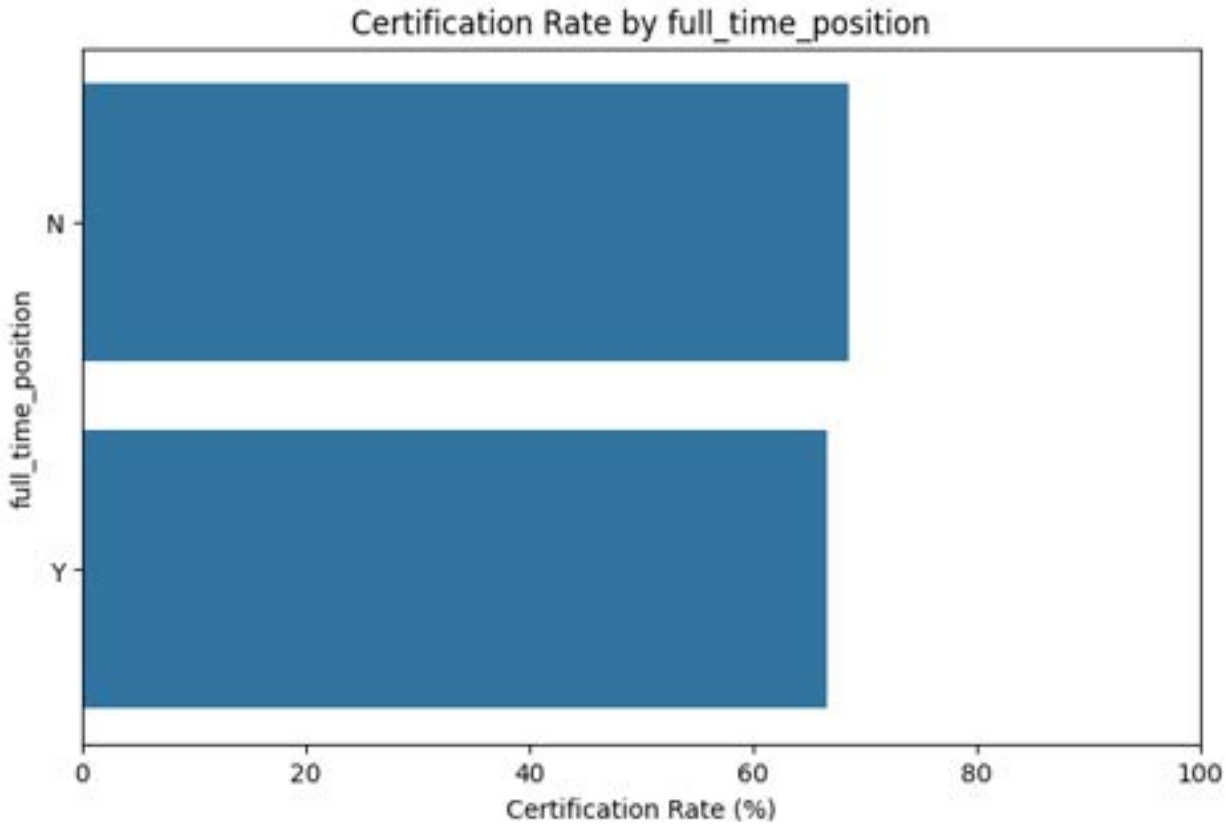

Certification Rate by education_of_employee
Doctorate
Master's
Bachelor's
High School
education_of_employee
0
20
40
60
80
100
Certification Rate (%)

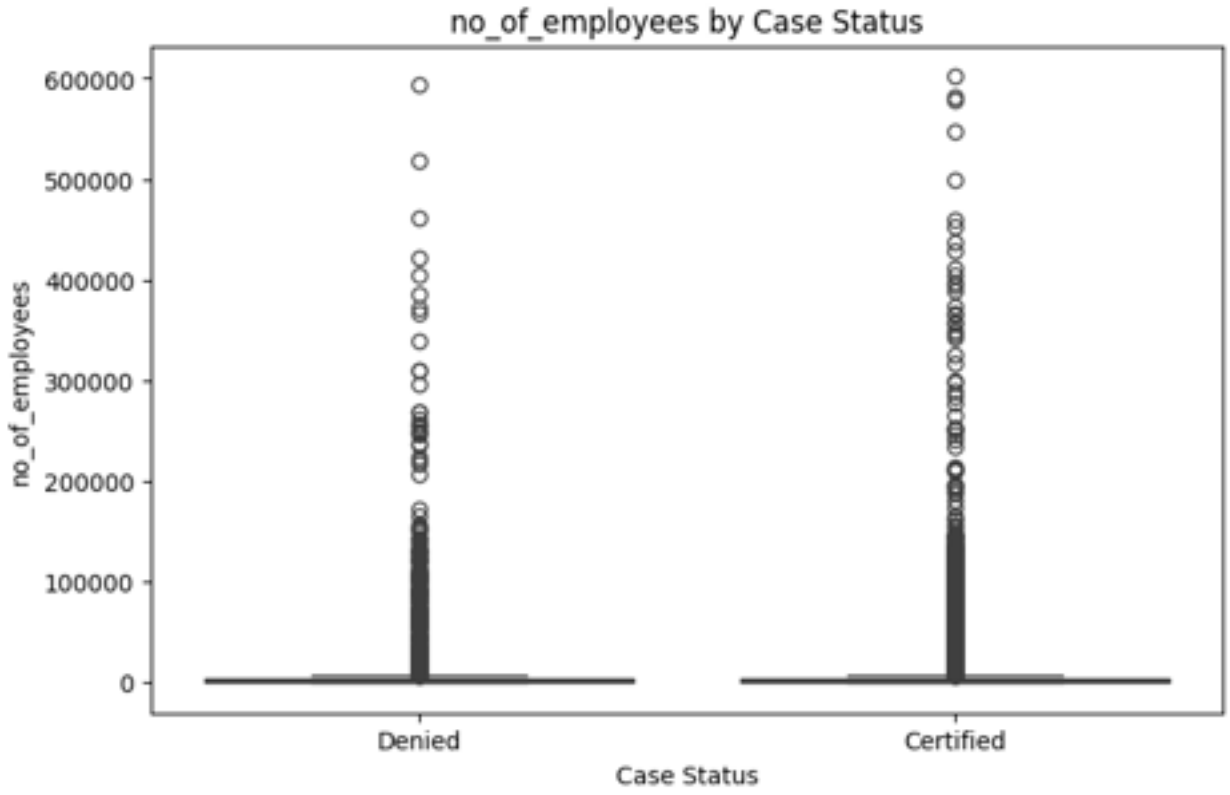

Certification Rate by has_job_experience
has_job_experience
Y
N
0
20
40
60
80
100
Certification Rate (%)


Certification Rate by requires_job_training
requires_job_training
Y
N
0
20
40
60
80
100
Certification Rate (%)

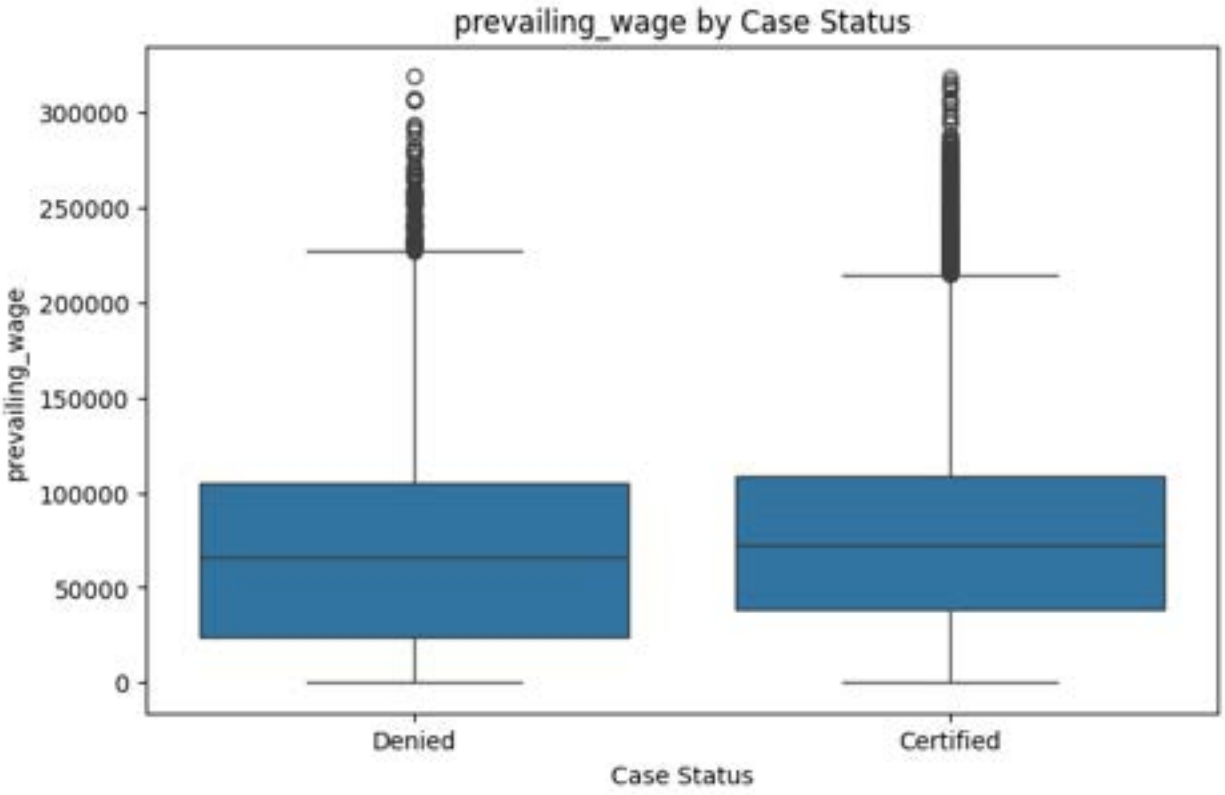

Certification Rate by region_of_employment
region_of_employment
Midwest
South
Northeast
West
Island
0
20
40
60
80
100
Certification Rate (%)


Certification Rate by unit_of_wage
unit_of_wage
Year
Week
Month
Hour
0
20
40
60
80
100
Certification Rate (%)


Certification Rate by full_time_position
full_time_position
N
Y
0
20
40
60
80
100
Certification Rate (%)


no_of_employees by Case Status
600000
500000
400000
300000
200000
100000
0
no_of_employees
Denied
Certified
Case Status


yr_of_estab by Case Status
2000
1950
1900
1850
1800
yr_of_estab
Denied
Certified
Case Status


prevailing_wage by Case Status
prevailing_wage
300000
250000
200000
150000
100000
50000
0
Denied
Certified
Case Status

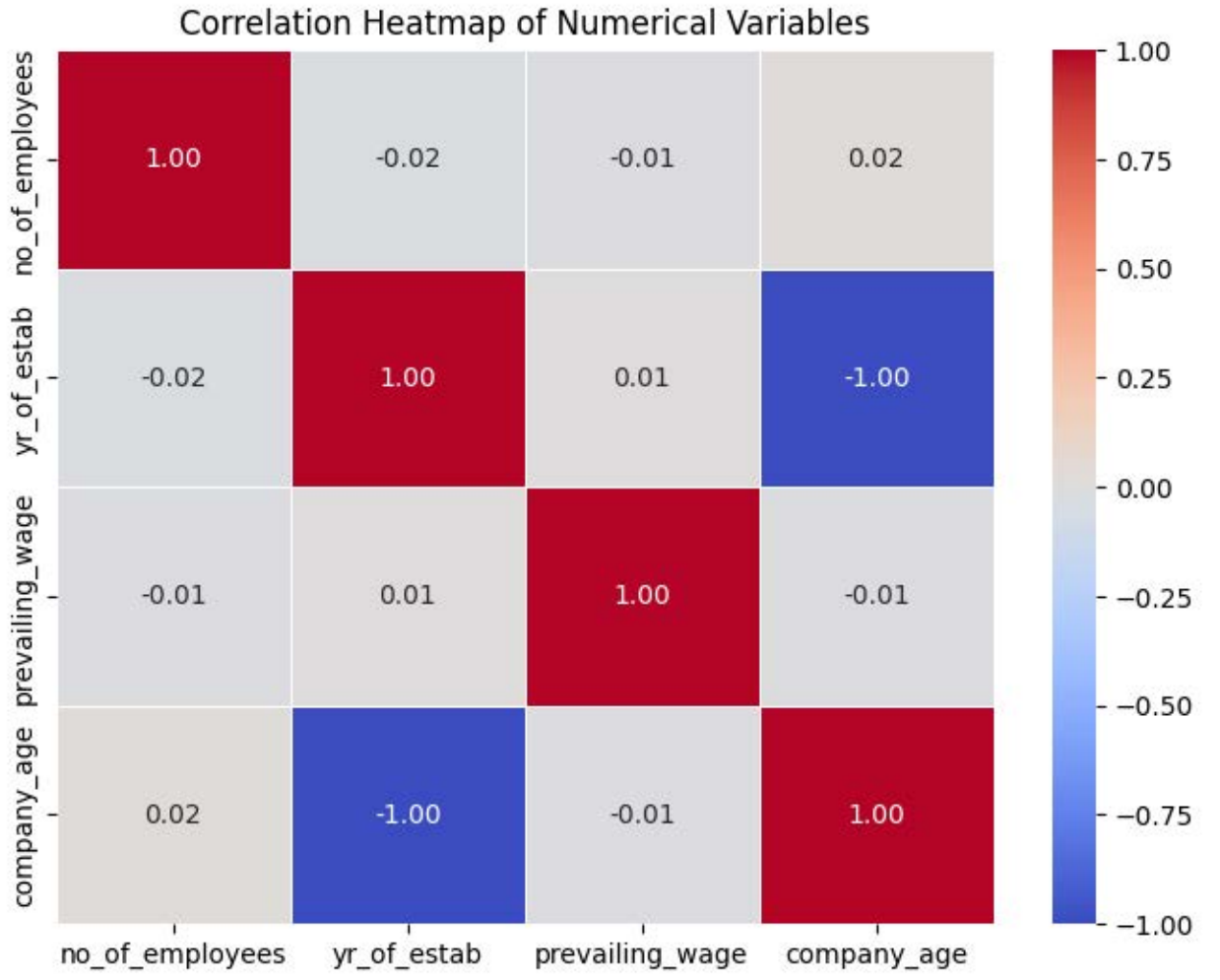

company_age by Case Status
company_age
200
150
100
50
0
Denied
Certified
Case Status

## 4.3 Correlation Analysis

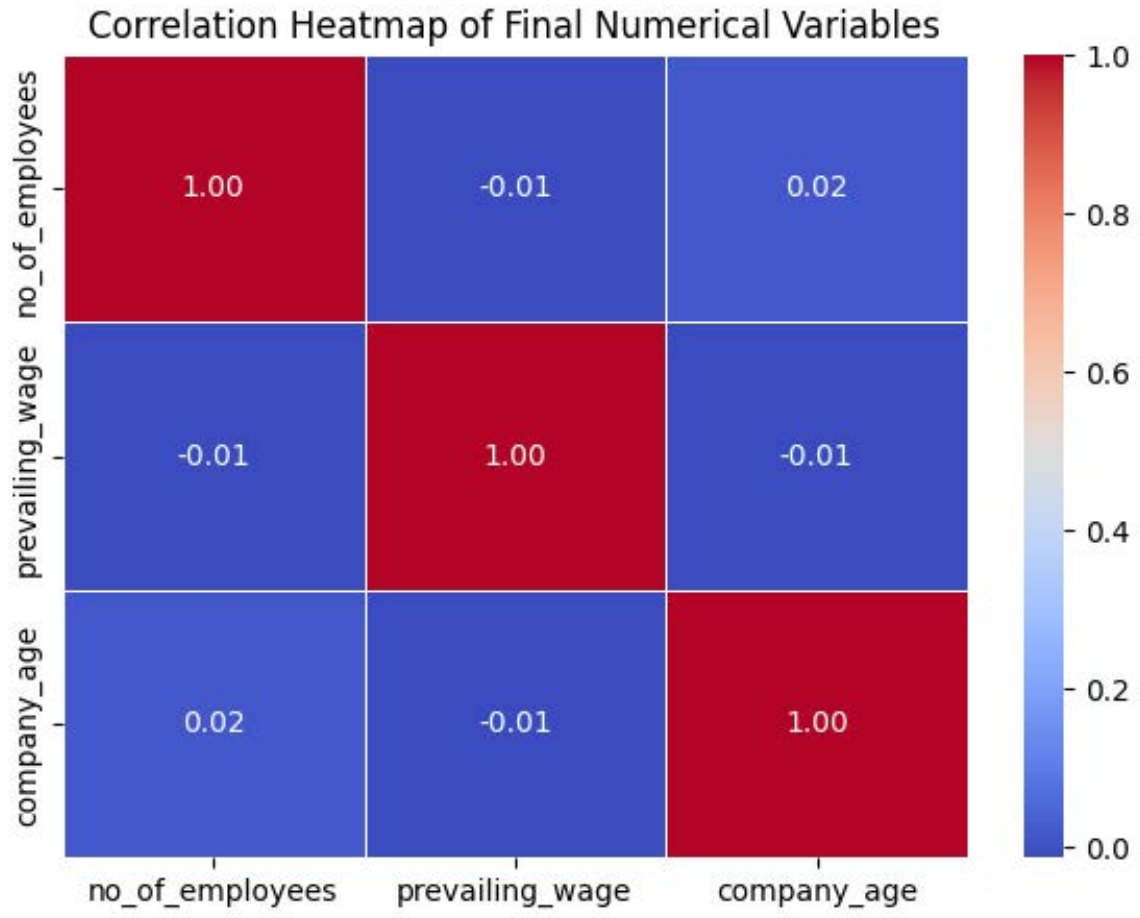
Correlation analysis was performed on numerical variables to identify relationships and potential multicollinearity.

The heatmap showed moderate relationships among selected numerical variables. No severe multicollinearity issues were observed.

| | no_of_employees | yr_of_estab | prevailing_wage | company_age |
|---|---|---|---|---|
| no_of_employees | 1.000000 | -0.017770 | -0.009523 | 0.017770 |
| yr_of_estab | -0.017770 | 1.000000 | 0.012342 | -1.000000 |
| prevailing_wage | -0.009523 | 0.012342 | 1.000000 | -0.012342 |
| company_age | 0.017770 | -1.000000 | -0.012342 | 1.000000 |



| | no_of_employees | prevailing_wage | company_age |
|---|---|---|---|
| **no_of_employees** | 1.000000 | -0.009523 | 0.017770 |
| **prevailing_wage** | -0.009523 | 1.000000 | -0.012342 |
| **company_age** | 0.017770 | -0.012342 | 1.000000 |



### 4.3 Correlation Analysis Observation

Correlation analysis was performed on numerical variables to identify relationships and potential multicollinearity.

The initial heatmap showed a perfect negative correlation between yr_of_estab and company_age, which is expected because company_age was directly created from yr_of_estab.

To avoid multicollinearity, yr_of_estab was dropped and company_age was retained for modeling because it is easier to interpret.

The final heatmap using no_of_employees, prevailing_wage, and company_age showed only weak to moderate relationships among the selected numerical variables.

No severe multicollinearity issues were observed among the final numerical predictors.

### Exploratory Data Analysis Summary

The EDA revealed the following important patterns:

- Certified applications were more common than denied applications.

- The target variable was moderately imbalanced.

- Higher education levels were associated with higher certification rates.

- Applicants with previous job experience had higher certification rates.

- Hourly wage applications had a much higher denial rate than yearly wage applications.

- Certified applications generally had higher prevailing wages.

- Certification rates varied by continent and region of employment.

- `no_of_employees` contained invalid negative values and extreme outliers.

- `case_id` was identified as an ID column and was not useful for prediction.

These insights helped guide preprocessing, feature engineering, and model selection.

## 5. Data Preprocessing

Several preprocessing techniques were applied to prepare the dataset for Machine Learning modeling.

The preprocessing workflow included:

- Creating a new feature called company_age using yr_of_estab
- Replacing invalid negative values in no_of_employees
- Removing case_id because it served only as an identifier
- Dropping yr_of_estab after feature engineering
- Encoding the target variable:
    - Certified = 1
    - Denied = 0
- Splitting data into training, validation, and test datasets
- Using stratified splitting to preserve class distribution
- Handling outliers using thresholds derived only from training data
- Applying median imputation for numerical variables
- Applying mode imputation for categorical variables
- Performing one-hot encoding for categorical variables

To prevent data leakage, all preprocessing operations were fitted only on training data and later applied to validation and test datasets.

Training shape: (17836, 10)

Validation shape: (3822, 10)

Test shape: (3822, 10)

Processed training shape: (17836, 28)

Processed validation shape: (3822, 28)

Processed test shape: (3822, 28)

| | **19643** | **17663** | **14342** | **8854** | **7811** |
|---|---|---|---|---|---|
| **no_of_employees** | 893.00 | 1850.00 | 1475.00 | 2877.00 | 523.00 |
| **prevailing_wage** | 141964.49 | 127273.02 | 77724.07 | 60777.57 | 50633.23 |
| **company_age** | 20.00 | 17.00 | 17.00 | 41.00 | 13.00 |
| **continent_Africa** | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| **continent_Asia** | 0.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| **continent_Europe** | 1.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| **continent_North America** | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| **continent_Oceania** | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| **continent_South America** | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| **education_of_employee_Bachelor's** | 0.00 | 0.00 | 1.00 | 0.00 | 1.00 |
| **education_of_employee_Doctorate** | 0.00 | 1.00 | 0.00 | 0.00 | 0.00 |
| **education_of_employee_High School** | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |

| | 19643 | 17663 | 14342 | 8854 | 7811 |
|---|---|---|---|---|---|
| education_of_employee_Master's | 1.00 | 0.00 | 0.00 | 1.00 | 0.00 |
| has_job_experience_N | 1.00 | 1.00 | 0.00 | 0.00 | 0.00 |
| has_job_experience_Y | 0.00 | 0.00 | 1.00 | 1.00 | 1.00 |
| requires_job_training_N | 0.00 | 0.00 | 1.00 | 1.00 | 1.00 |
| requires_job_training_Y | 1.00 | 1.00 | 0.00 | 0.00 | 0.00 |
| region_of_employment_Island | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| region_of_employment_Midwest | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| region_of_employment_Northeast | 1.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| region_of_employment_South | 0.00 | 1.00 | 1.00 | 1.00 | 0.00 |
| region_of_employment_West | 0.00 | 0.00 | 0.00 | 0.00 | 1.00 |
| unit_of_wage_Hour | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| unit_of_wage_Month | 1.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| unit_of_wage_Week | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| unit_of_wage_Year | 0.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| full_time_position_N | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| full_time_position_Y | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |

## Data Preprocessing Summary

The following preprocessing steps were performed:

- Created a new feature called `company_age` from `yr_of_estab`
- Replaced invalid negative values in `no_of_employees`
- Dropped `case_id` because it is only an identifier
- Dropped `yr_of_estab` after creating `company_age`
- Encoded the target variable where Certified = 1 and Denied = 0
- Split the data into training, validation, and test sets
- Used stratified splitting to preserve target distribution
- Treated outliers using limits calculated only from training data
- Imputed numerical values using median imputation
- Imputed categorical values using most frequent category
- One-hot encoded categorical variables
- Fitted preprocessing only on training data to avoid data leakage

These steps ensured the data was ready for model building and evaluation.

## 6. Model Building and Evaluation

Machine Learning models were developed using:

1. Original data
2. Oversampled data
3. Undersampled data

The following classification algorithms were implemented:

- Decision Tree
- Bagging Classifier

- Random Forest
- Extra Trees
- AdaBoost
- Gradient Boosting

F1-score was selected as the primary evaluation metric because it balances precision and recall while addressing moderate class imbalance.

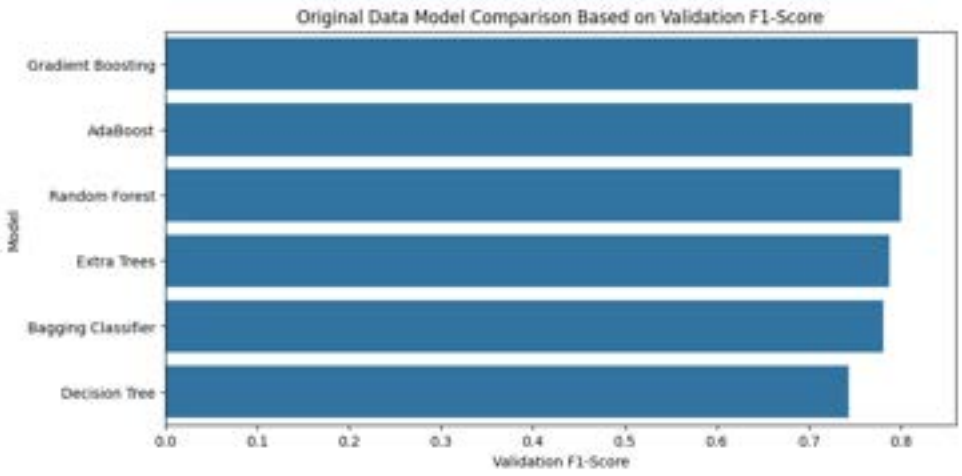
Additional evaluation metrics included:

- Accuracy
- Precision
- Recall
- ROC-AUC
- Confusion Matrix

```
case_status
1    66.791882
0    33.208118
Name: proportion, dtype: float64
Since the target variable is moderately imbalanced, F1-score will be used as an important evaluation metric.
```

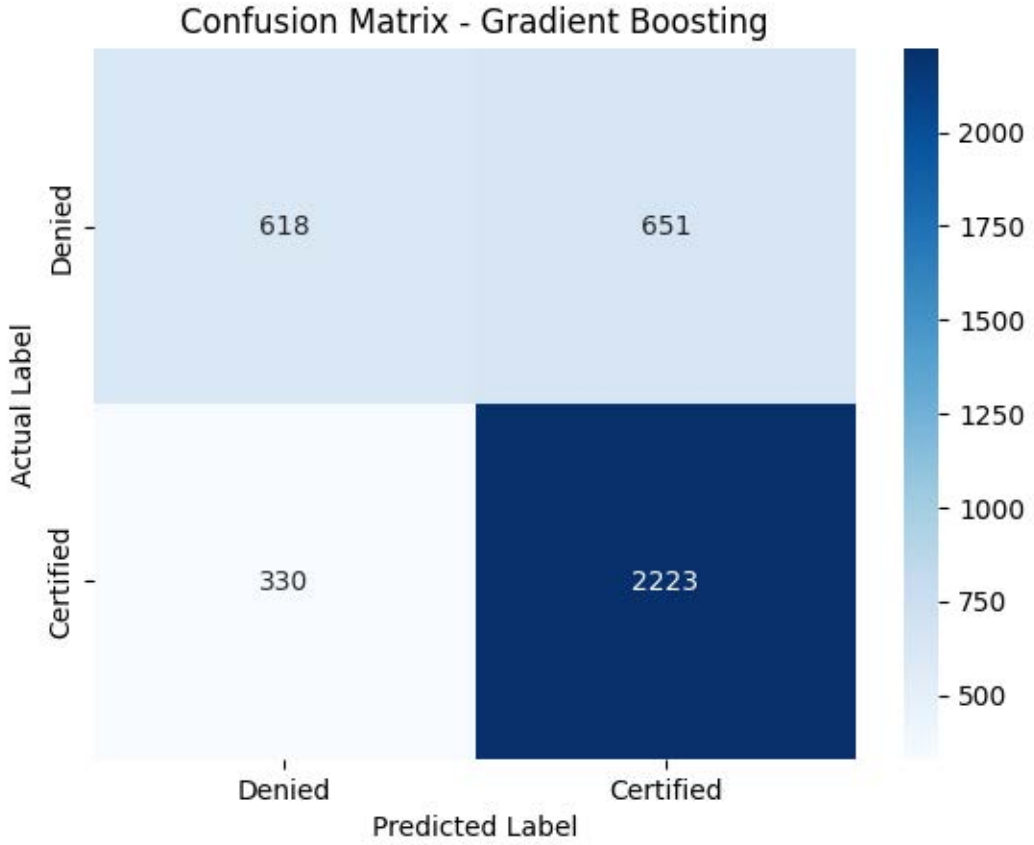
| | **5** | **4** | **2** | **3** | **1** | **0** |
|---|---|---|---|---|---|---|
| **Model** | Gradient Boosting | AdaBoost | Random Forest | Extra Trees | Bagging Classifier | Decision Tree |
| **Train Accuracy** | 0.75712 | 0.738899 | 1.0 | 1.0 | 0.984862 | 1.0 |
| **Validation Accuracy** | 0.743328 | 0.729723 | 0.722135 | 0.70539 | 0.704605 | 0.657509 |

| | 5 | 4 | 2 | 3 | 1 | 0 |
|---|---|---|---|---|---|---|
| **Train Precision** | 0.784252 | 0.762594 | 1.0 | 1.0 | 0.992221 | 1.0 |
| **Validation Precision** | 0.773486 | 0.755892 | 0.768455 | 0.757488 | 0.771548 | 0.74373 |
| **Train Recall** | 0.877865 | 0.884412 | 1.0 | 1.0 | 0.985058 | 1.0 |
| **Validation Recall** | 0.87074 | 0.879358 | 0.835879 | 0.82217 | 0.792401 | 0.743439 |
| **Train F1** | 0.828422 | 0.818998 | 1.0 | 1.0 | 0.988627 | 1.0 |
| **Validation F1** | 0.819237 | 0.812964 | 0.80075 | 0.788505 | 0.781836 | 0.743585 |
| **Train ROC-AUC** | 0.796494 | 0.773265 | 1.0 | 1.0 | 0.998554 | 1.0 |
| **Validation ROC-AUC** | 0.779194 | 0.76292 | 0.752953 | 0.742291 | 0.725527 | 0.614036 |



Best model on original data: Gradient Boosting

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| Denied | 0.65 | 0.49 | 0.56 | 1269 |
| Certified | 0.77 | 0.87 | 0.82 | 2553 |
| accuracy | | | 0.74 | 3822 |
| macro avg | 0.71 | 0.68 | 0.69 | 3822 |
| weighted avg | 0.73 | 0.74 | 0.73 | 3822 |



## 6.2 Oversampled Data Models

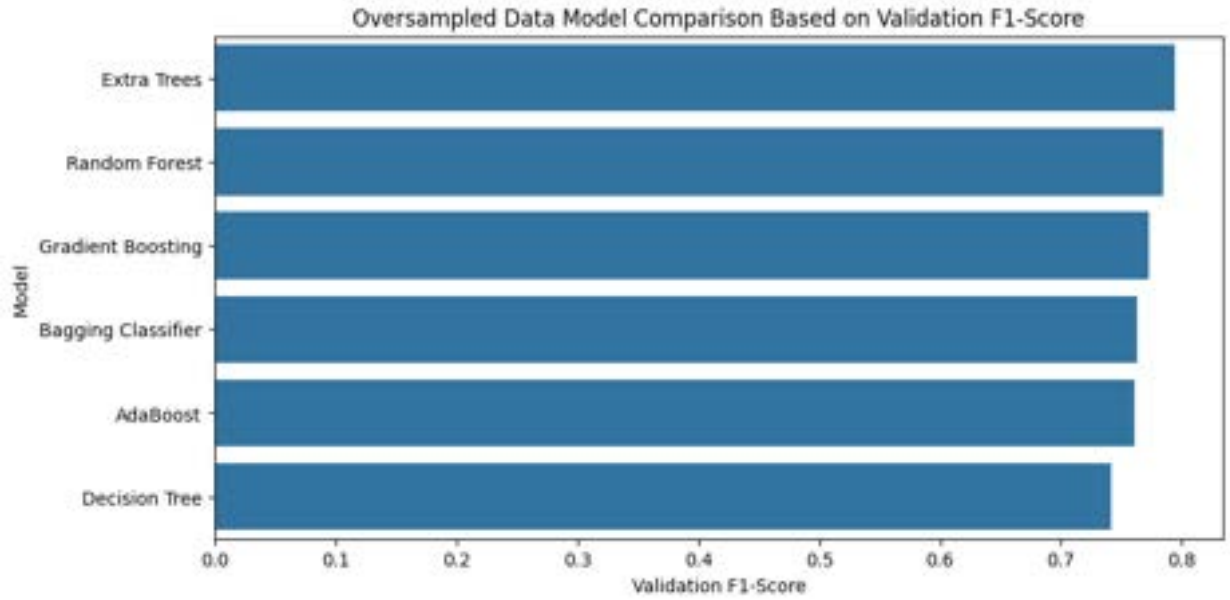
SMOTE oversampling was applied to improve minority class representation.

Oversampling generally improved recall performance and enhanced minority class prediction capability.

```
Before oversampling:
case_status
1    11913
0     5923
Name: count, dtype: int64

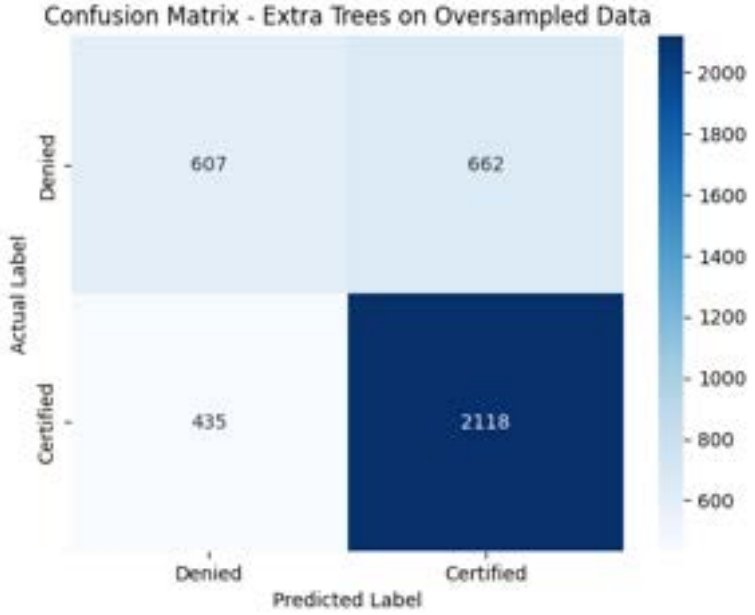
Oversampled X_train shape: (23826, 28)
Oversampled y_train shape: (23826,)
After oversampling:
case_status
0    11913
1    11913
Name: count, dtype: int64
```

| | 3 | 2 | 5 | 1 | 4 | 0 |
|---|---|---|---|---|---|---|
| **Model** | Extra Trees | Random Forest | Gradient Boosting | Bagging Classifier | AdaBoost | Decision Tree |
| **Train Accuracy** | 1.0 | 0.999958 | 0.721061 | 0.988416 | 0.693948 | 1.0 |
| **Validation Accuracy** | 0.712977 | 0.709053 | 0.710623 | 0.692308 | 0.694662 | 0.653061 |
| **Train Precision** | 1.0 | 1.0 | 0.711849 | 0.99701 | 0.680437 | 1.0 |
| **Validation Precision** | 0.761871 | 0.776796 | 0.813883 | 0.786041 | 0.798193 | 0.738438 |
| **Train Recall** | 1.0 | 0.999916 | 0.742802 | 0.97977 | 0.731386 | 1.0 |
| **Validation Recall** | 0.829612 | 0.792009 | 0.734822 | 0.741089 | 0.726596 | 0.744222 |
| **Train F1** | 1.0 | 0.999958 | 0.726996 | 0.988315 | 0.704992 | 1.0 |
| **Validation F1** | 0.7943 | 0.784329 | 0.772334 | 0.762903 | 0.760714 | 0.741319 |
| **Train ROC-AUC** | 1.0 | 1.0 | 0.797774 | 0.999595 | 0.770363 | 1.0 |
| **Validation ROC-AUC** | 0.744761 | 0.749539 | 0.779141 | 0.733659 | 0.76238 | 0.606942 |




Best model on oversampled data: Extra Trees

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| Denied | 0.58 | 0.48 | 0.53 | 1269 |
| Certified | 0.76 | 0.83 | 0.79 | 2553 |
| accuracy | | | 0.71 | 3822 |
| macro avg | 0.67 | 0.65 | 0.66 | 3822 |
| weighted avg | 0.70 | 0.71 | 0.70 | 3822 |



| | 3 | 2 | 5 | 1 | 4 | 0 |
|---|---|---|---|---|---|---|
| **Model** | Extra Trees | Random Forest | Gradient Boosting | Bagging Classifier | AdaBoost | Decision Tree |
| **Train Accuracy** | 1.0 | 0.999958 | 0.721061 | 0.988416 | 0.693948 | 1.0 |
| **Validation Accuracy** | 0.712977 | 0.709053 | 0.710623 | 0.692308 | 0.694662 | 0.653061 |
| **Train Precision** | 1.0 | 1.0 | 0.711849 | 0.99701 | 0.680437 | 1.0 |
| **Validation Precision** | 0.761871 | 0.776796 | 0.813883 | 0.786041 | 0.798193 | 0.738438 |
| **Train Recall** | 1.0 | 0.999916 | 0.742802 | 0.97977 | 0.731386 | 1.0 |
| **Validation Recall** | 0.829612 | 0.792009 | 0.734822 | 0.741089 | 0.726596 | 0.744222 |
| **Train F1** | 1.0 | 0.999958 | 0.726996 | 0.988315 | 0.704992 | 1.0 |
| **Validation F1** | 0.7943 | 0.784329 | 0.772334 | 0.762903 | 0.760714 | 0.741319 |

| | 3 | 2 | 5 | 1 | 4 | 0 |
|---|---|---|---|---|---|---|
| **Train ROC-AUC** | 1.0 | 1.0 | 0.797774 | 0.999595 | 0.770363 | 1.0 |
| **Validation ROC-AUC** | 0.744761 | 0.749539 | 0.779141 | 0.733659 | 0.76238 | 0.606942 |
| **Data Type** | Oversampled Data | Oversampled Data | Oversampled Data | Oversampled Data | Oversampled Data | Oversampled Data |

### Summary

By using oversampling, we make sure that the training dataset is balanced between Certified and Denied class. It might be helpful in making the model understand the minority class better.

After oversampling, the minority class recall might improve due to increased examples of the denied applications in the training data. However, since we duplicate minority class observations during oversampling, we might increase the risk of overfitting, especially for tree-based models.

Selection criteria for the best model after oversampling include mostly validation F1 score, however, validation recall, precision and ROC-AUC must be considered as well.

## 6.3 Undersampled Data Models

Undersampling was performed to balance the dataset by reducing majority class samples.

Undersampled models trained faster but risked losing valuable information from removed observations.

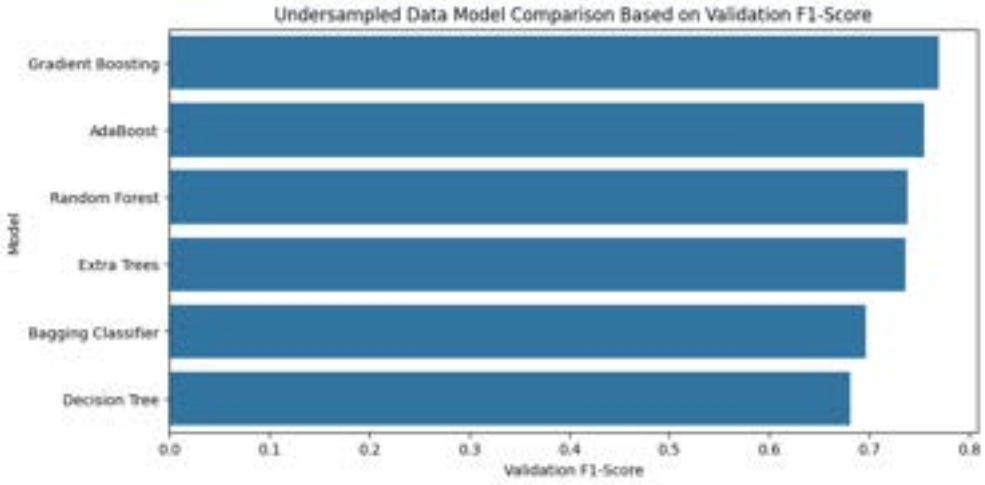
```
Before undersampling:
case_status
1    11913
0     5923
Name: count, dtype: int64

Undersampled X_train shape: (11846, 28)
Undersampled y_train shape: (11846,)

After undersampling:
case_status
0    5923
1    5923
Name: count, dtype: int64
```

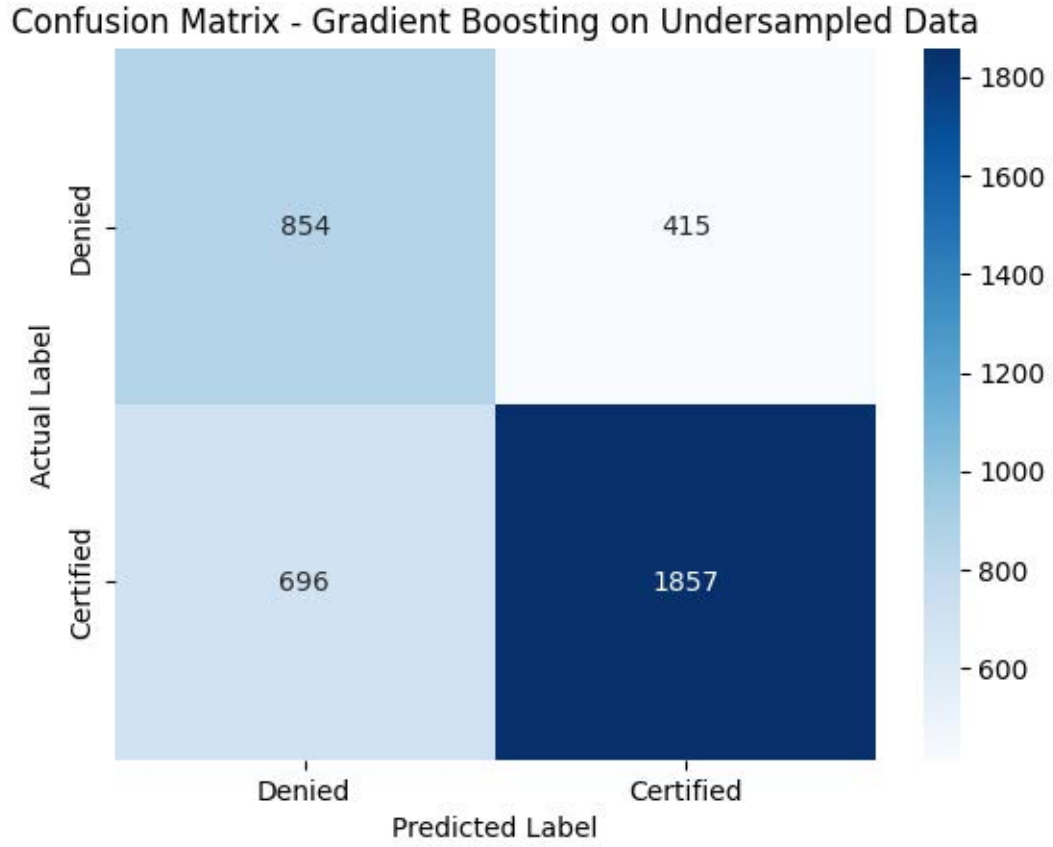
| | **5** | **4** | **2** | **3** | **1** | **0** |
|---|---|---|---|---|---|---|
| **Model** | Gradient Boosting | AdaBoost | Random Forest | Extra Trees | Bagging Classifier | Decision Tree |
| **Train Accuracy** | 0.721087 | 0.695256 | 1.0 | 1.0 | 0.978389 | 1.0 |
| **Validation Accuracy** | 0.709314 | 0.689691 | 0.677656 | 0.673469 | 0.641287 | 0.616431 |
| **Train Precision** | 0.714637 | 0.686022 | 1.0 | 1.0 | 0.99116 | 1.0 |
| **Validation Precision** | 0.817342 | 0.799912 | 0.807066 | 0.799175 | 0.801223 | 0.76577 |
| **Train Recall** | 0.736113 | 0.720074 | 1.0 | 1.0 | 0.965389 | 1.0 |

| | 5 | 4 | 2 | 3 | 1 | 0 |
|---|---|---|---|---|---|---|
| **Validation Recall** | 0.72738 | 0.714062 | 0.679984 | 0.682726 | 0.615746 | 0.613396 |
| **Train F1** | 0.725216 | 0.702636 | 1.0 | 1.0 | 0.978105 | 1.0 |
| **Validation F1** | 0.769741 | 0.754553 | 0.738095 | 0.736375 | 0.696346 | 0.681166 |
| **Train ROC-AUC** | 0.79718 | 0.772234 | 1.0 | 1.0 | 0.998301 | 1.0 |
| **Validation ROC-AUC** | 0.776546 | 0.76211 | 0.749779 | 0.740868 | 0.728808 | 0.617967 |



Best model on undersampled data: Gradient Boosting

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| Denied | 0.55 | 0.67 | 0.61 | 1269 |
| Certified | 0.82 | 0.73 | 0.77 | 2553 |
| accuracy | | | 0.71 | 3822 |
| macro avg | 0.68 | 0.70 | 0.69 | 3822 |
| weighted avg | 0.73 | 0.71 | 0.72 | 3822 |



| | 5 | 4 | 2 | 3 | 1 | 0 |
|---|---|---|---|---|---|---|
| **Model** | Gradient Boosting | AdaBoost | Random Forest | Extra Trees | Bagging Classifier | Decision Tree |
| **Train Accuracy** | 0.721087 | 0.695256 | 1.0 | 1.0 | 0.978389 | 1.0 |
| **Validation Accuracy** | 0.709314 | 0.689691 | 0.677656 | 0.673469 | 0.641287 | 0.616431 |
| **Train Precision** | 0.714637 | 0.686022 | 1.0 | 1.0 | 0.99116 | 1.0 |
| **Validation Precision** | 0.817342 | 0.799912 | 0.807066 | 0.799175 | 0.801223 | 0.76577 |
| **Train Recall** | 0.736113 | 0.720074 | 1.0 | 1.0 | 0.965389 | 1.0 |
| **Validation Recall** | 0.72738 | 0.714062 | 0.679984 | 0.682726 | 0.615746 | 0.613396 |
| **Train F1** | 0.725216 | 0.702636 | 1.0 | 1.0 | 0.978105 | 1.0 |
| **Validation F1** | 0.769741 | 0.754553 | 0.738095 | 0.736375 | 0.696346 | 0.681166 |

| | 5 | 4 | 2 | 3 | 1 | 0 |
|---|---|---|---|---|---|---|
| **Train ROC-AUC** | 0.79718 | 0.772234 | 1.0 | 1.0 | 0.998301 | 1.0 |
| **Validation ROC-AUC** | 0.776546 | 0.76211 | 0.749779 | 0.740868 | 0.728808 | 0.617967 |
| **Data Type** | Undersampled Data | Undersampled Data | Undersampled Data | Undersampled Data | Undersampled Data | Undersampled Data |

**Summary**

Under-sampling will balance both classes by decreasing the numbers in Certified category in the training dataset.

This will possibly help the model focus more on the Denied category. But since under-sampling will discard the majority class, it may lead to loss of relevant information.

That is why undersampled models can enhance recall in terms of denied applications but may sacrifice accuracy and stability.

The optimal undersampled model needs to be selected based on validation F1 score as well as other metrics like recall, precision, and ROC-AUC.

## 7. Model Performance Improvement using Hyperparameter Tuning.

Hyperparameter tuning was performed using RandomizedSearchCV to improve predictive performance efficiently.

The following models were selected for tuning:

- Random Forest
- Extra Trees
- Gradient Boosting

These ensemble models were selected because they generally perform well on structured tabular data and effectively capture complex non-linear relationships.

F1-score was used as the tuning metric.

F1-score is used as the tuning metric because it balances precision and recall.

RandomizedSearchCV will be used to improve selected model performance.

Fitting 3 folds for each of 20 candidates, totaling 60 fits

### 7.1 Hyperparameter Search Space

Best Random Forest Parameters: {'n_estimators': 200, 'min_samples_split': 5, 'min_samples_leaf': 4, 'max_features': 'log2', 'max_depth': 10}

Best Random Forest CV F1-score: 0.8252360506177171

Fitting 3 folds for each of 20 candidates, totalling 60 fits

Best Extra Trees Parameters: {'n_estimators': 300, 'min_samples_split': 10, 'min_samples_leaf': 2, 'max_features': 'log2', 'max_depth': 10}

Best Extra Trees CV F1-score: 0.8257037784617328

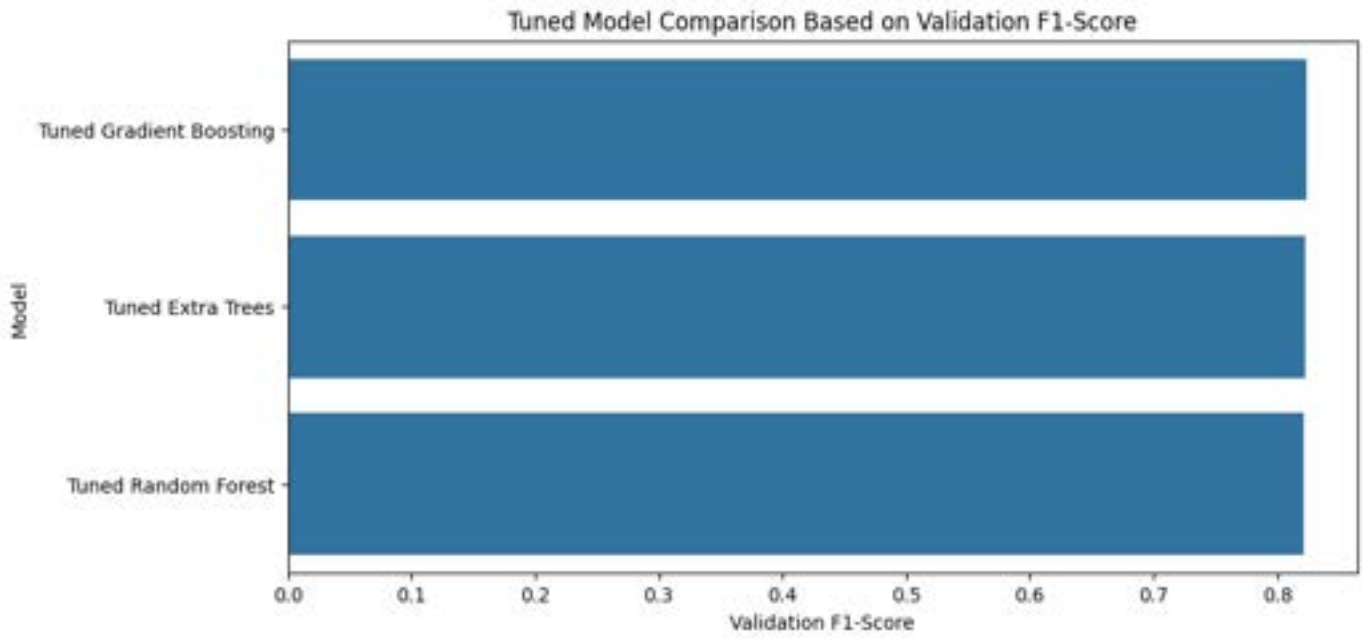
Fitting 3 folds for each of 20 candidates, totalling 60 fits

Best Gradient Boosting Parameters: {'subsample': 0.8, 'n_estimators': 150, 'min_samples_split': 5, 'min_samples_leaf': 4, 'max_depth': 5, 'learning_rate': 0.01}

Best Gradient Boosting CV F1-score: 0.8248136765385734

| | **2** | **1** | **0** |
|---|---|---|---|
| **Model** | Tuned Gradient Boosting | Tuned Extra Trees | Tuned Random Forest |
| **Train Accuracy** | 0.748542 | 0.761045 | 0.764297 |
| **Validation Accuracy** | 0.736525 | 0.745683 | 0.74516 |
| **Train Precision** | 0.754698 | 0.783938 | 0.785075 |
| **Validation Precision** | 0.745086 | 0.77137 | 0.770284 |
| **Train Recall** | 0.923781 | 0.886594 | 0.891043 |
| **Validation Recall** | 0.920486 | 0.880141 | 0.881316 |
| **Train F1** | 0.830723 | 0.832112 | 0.834709 |
| **Validation F1** | 0.82355 | 0.822173 | 0.822068 |
| **Train ROC-AUC** | 0.792702 | 0.812667 | 0.820838 |

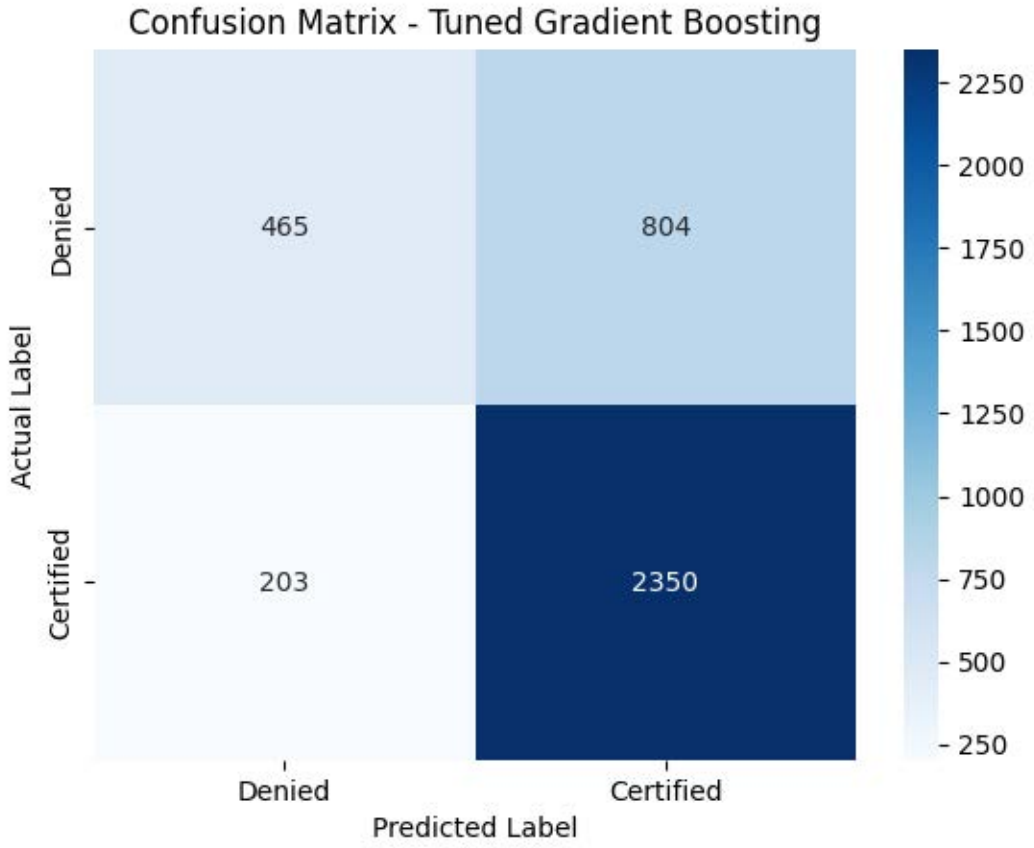
| | 2 | 1 | 0 |
|---|---|---|---|
| **Validation ROC-AUC** | 0.774292 | 0.781249 | 0.780488 |




Best tuned model: Tuned Gradient Boosting

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| Denied | 0.70 | 0.37 | 0.48 | 1269 |
| Certified | 0.75 | 0.92 | 0.82 | 2553 |
| accuracy | | | 0.74 | 3822 |
| macro avg | 0.72 | 0.64 | 0.65 | 3822 |

| | | | | |
|---|---|---|---|---|
| weighted avg | 0.73 | 0.74 | 0.71 | 3822 |



| | **2** | **1** | **0** |
|---|---|---|---|
| **Model** | Tuned Gradient Boosting | Tuned Extra Trees | Tuned Random Forest |
| **Train Accuracy** | 0.748542 | 0.761045 | 0.764297 |
| **Validation Accuracy** | 0.736525 | 0.745683 | 0.74516 |
| **Train Precision** | 0.754698 | 0.783938 | 0.785075 |
| **Validation Precision** | 0.745086 | 0.77137 | 0.770284 |
| **Train Recall** | 0.923781 | 0.886594 | 0.891043 |
| **Validation Recall** | 0.920486 | 0.880141 | 0.881316 |
| **Train F1** | 0.830723 | 0.832112 | 0.834709 |
| **Validation F1** | 0.82355 | 0.822173 | 0.822068 |
| **Train ROC-AUC** | 0.792702 | 0.812667 | 0.820838 |
| **Validation ROC-AUC** | 0.774292 | 0.781249 | 0.780488 |
| **Data Type** | Tuned Original Data | Tuned Original Data | Tuned Original Data |

## 8. Final Model Selection

After comparing models trained on original, oversampled, undersampled, and tuned datasets, the final selected model was chosen based on:

- F1-score
- Precision
- Recall
- ROC-AUC
- Overall generalization performance

| | 18 | 19 | 20 | 0 | 1 | 2 | 6 | 3 | 7 | 4 | ... | 12 | 9 | 10 | 13 | 5 | 11 | 14 | 15 | 16 | 17 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Model** | Tuned Gradient Boosting | Tuned Extra Trees | Tuned Random Forest | Gradient Boosting | AdaBoost | Random Forest | Extra Trees | Extra Trees | Random Forest | Bagging Classifier | ... | Gradient Boosting | Bagging Classifier | AdaBoost | AdaBoost | Decision Tree | Decision Tree | Random Forest | Extra Trees | Bagging Classifier | Decision Tree |

| | 18 | 19 | 20 | 0 | 1 | 2 | 6 | 3 | 7 | 4 | ... | 12 | 9 | 10 | 13 | 5 | 11 | 14 | 15 | 16 | 17 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Train Accuracy** | 0.748542 | 0.761045 | 0.764297 | 0.757112 | 0.738899 | 1.0 | 1.0 | 1.0 | 0.999958 | 0.984862 | ... | 0.721087 | 0.988416 | 0.693948 | 0.695256 | 1.0 | 1.0 | 1.0 | 1.0 | 0.978389 | 1.0 |
| **Validation Accuracy** | 0.736525 | 0.745683 | 0.745116 | 0.743328 | 0.722973 | 0.722135 | 0.712977 | 0.705339 | 0.709053 | 0.704605 | ... | 0.709314 | 0.692308 | 0.694662 | 0.689691 | 0.657509 | 0.653061 | 0.677656 | 0.673469 | 0.641287 | 0.616431 |
| **Train Precision** | 0.754698 | 0.783938 | 0.785075 | 0.784252 | 0.762594 | 1.0 | 1.0 | 1.0 | 1.0 | 0.992221 | ... | 0.714637 | 0.997001 | 0.680437 | 0.686022 | 1.0 | 1.0 | 1.0 | 1.0 | 0.991116 | 1.0 |
| **Validation** | 0.7745 | 0.7771 | 0.7770 | 0.7773 | 0.7755 | 0.7768 | 0.761871 | 0.7757 | 0.776796 | 0.7771 | ... | 0.817342 | 0.786041 | 0.798193 | 0.799912 | 0.7443 | 0.738438 | 0.807066 | 0.799175 | 0.801223 | 0.767577 |

| | 18 | 19 | 20 | 0 | 1 | 2 | 6 | 3 | 7 | 4 | ... | 12 | 9 | 10 | 13 | 5 | 11 | 14 | 15 | 16 | 17 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Precision** | 086 | 37 | 284 | 486 | 892 | 455 | | 488 | | 548 | | | | | | 73 | | | | | |
| **Train Recall** | 0.923781 | 0.886594 | 0.891043 | 0.877865 | 0.884412 | 1.0 | 1.0 | 1.0 | 0.999916 | 0.985058 | ... | 0.736113 | 0.97977 | 0.731386 | 0.720074 | 1.0 | 1.0 | 1.0 | 1.0 | 0.965389 | 1.0 |
| **Validation Recall** | 0.920486 | 0.880141 | 0.881316 | 0.870774 | 0.879358 | 0.835879 | 0.829612 | 0.822117 | 0.792009 | 0.792401 | ... | 0.72738 | 0.741089 | 0.726596 | 0.714062 | 0.743439 | 0.744222 | 0.679984 | 0.682726 | 0.615746 | 0.613396 |
| **Train F1** | 0.830723 | 0.832112 | 0.834709 | 0.828422 | 0.818998 | 1.0 | 1.0 | 1.0 | 0.999958 | 0.988627 | ... | 0.725216 | 0.988315 | 0.704992 | 0.702636 | 1.0 | 1.0 | 1.0 | 1.0 | 0.978105 | 1.0 |

| | 18 | 19 | 20 | 0 | 1 | 2 | 6 | 3 | 7 | 4 | ... | 12 | 9 | 10 | 13 | 5 | 11 | 14 | 15 | 16 | 17 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **ValidationF1** | 0.823555 | 0.822173 | 0.822068 | 0.819237 | 0.812964 | 0.800075 | 0.7943 | 0.788505 | 0.784329 | 0.781836 | ... | 0.769741 | 0.762903 | 0.760714 | 0.754553 | 0.743585 | 0.741319 | 0.738095 | 0.736375 | 0.696346 | 0.681166 |
| **TrainROC-AUC** | 0.792702 | 0.812667 | 0.820838 | 0.796494 | 0.773265 | 1.0 | 1.0 | 1.0 | 1.0 | 0.998554 | ... | 0.7971 8 | 0.999595 | 0.770363 | 0.772234 | 1.0 | 1.0 | 1.0 | 1.0 | 0.998301 | 1.0 |
| **ValidationROC-AUC** | 0.774292 | 0.781249 | 0.780488 | 0.779194 | 0.762992 | 0.752953 | 0.744761 | 0.742291 | 0.749539 | 0.725527 | ... | 0.765 46 | 0.733659 | 0.76238 | 0.762111 | 0.614036 | 0.606942 | 0.749779 | 0.740868 | 0.728808 | 0.617967 |

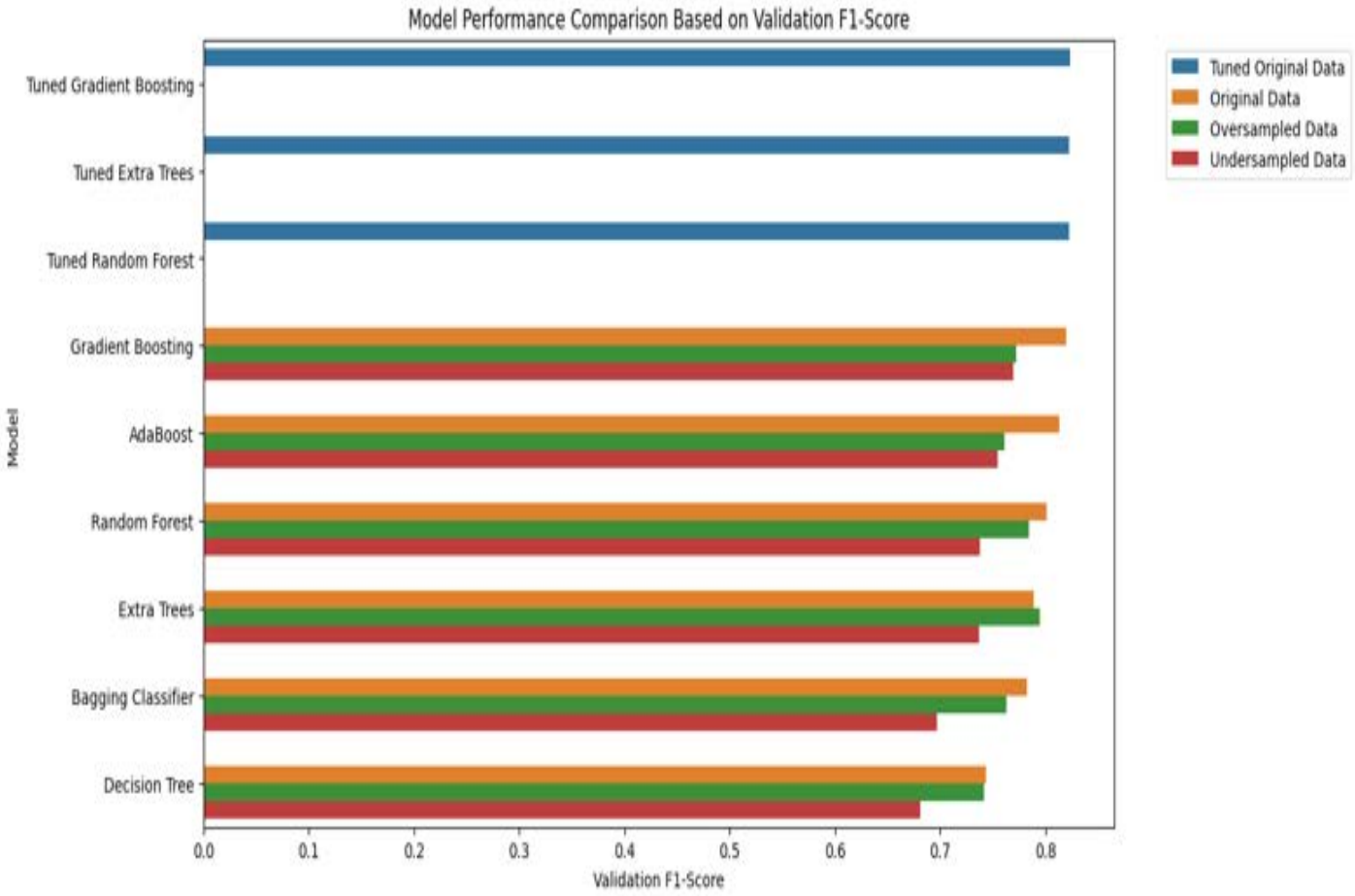
| | 18 | 19 | 20 | 0 | 1 | 2 | 6 | 3 | 7 | 4 | ... | 12 | 9 | 10 | 13 | 5 | 11 | 14 | 15 | 16 | 17 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Data Type** | Tuned Original Data | Tuned Original Data | Tuned Original Data | Original Data | Original Data | Original Data | Oversampled Data | Original Data | Oversampled Data | Original Data | ... | Undersampled Data | Oversampled Data | Oversampled Data | Undersampled Data | Original Data | Oversampled Data | Undersampled Data | Undersampled Data | Undersampled Data | Undersampled Data |

12 rows × 21 columns

| | Model | Data Type | Validation Accuracy | Validation Precision | Validation Recall | Validation F1 | Validation ROC-AUC |
|---|---|---|---|---|---|---|---|
| **18** | Tuned Gradient Boosting | Tuned Original Data | 0.736525 | 0.745086 | 0.920486 | 0.823550 | 0.774292 |
| **19** | Tuned Extra Trees | Tuned Original Data | 0.745683 | 0.771370 | 0.880141 | 0.822173 | 0.781249 |
| **20** | Tuned Random Forest | Tuned Original Data | 0.745160 | 0.770284 | 0.881316 | 0.822068 | 0.780488 |

| | Model | Data Type | Validation Accuracy | Validation Precision | Validation Recall | Validation F1 | Validation ROC-AUC |
|---|---|---|---|---|---|---|---|
| **0** | Gradient Boosting | Original Data | 0.743328 | 0.773486 | 0.870740 | 0.819237 | 0.779194 |
| **1** | AdaBoost | Original Data | 0.729723 | 0.755892 | 0.879358 | 0.812964 | 0.762920 |
| **2** | Random Forest | Original Data | 0.722135 | 0.768455 | 0.835879 | 0.800750 | 0.752953 |
| **6** | Extra Trees | Oversampled Data | 0.712977 | 0.761871 | 0.829612 | 0.794300 | 0.744761 |
| **3** | Extra Trees | Original Data | 0.705390 | 0.757488 | 0.822170 | 0.788505 | 0.742291 |
| **7** | Random Forest | Oversampled Data | 0.709053 | 0.776796 | 0.792009 | 0.784329 | 0.749539 |
| **4** | Bagging Classifier | Original Data | 0.704605 | 0.771548 | 0.792401 | 0.781836 | 0.725527 |
| **8** | Gradient Boosting | Oversampled Data | 0.710623 | 0.813883 | 0.734822 | 0.772334 | 0.779141 |
| **12** | Gradient Boosting | Undersampled Data | 0.709314 | 0.817342 | 0.727380 | 0.769741 | 0.776546 |
| **9** | Bagging Classifier | Oversampled Data | 0.692308 | 0.786041 | 0.741089 | 0.762903 | 0.733659 |
| **10** | AdaBoost | Oversampled Data | 0.694662 | 0.798193 | 0.726596 | 0.760714 | 0.762380 |

| | Model | Data Type | Validation Accuracy | Validation Precision | Validation Recall | Validation F1 | Validation ROC-AUC |
|---|---|---|---|---|---|---|---|
| **13** | AdaBoost | Undersampled Data | 0.689691 | 0.799912 | 0.714062 | 0.754553 | 0.762110 |
| **5** | Decision Tree | Original Data | 0.657509 | 0.743730 | 0.743439 | 0.743585 | 0.614036 |
| **11** | Decision Tree | Oversampled Data | 0.653061 | 0.738438 | 0.744222 | 0.741319 | 0.606942 |
| **14** | Random Forest | Undersampled Data | 0.677656 | 0.807066 | 0.679984 | 0.738095 | 0.749779 |
| **15** | Extra Trees | Undersampled Data | 0.673469 | 0.799175 | 0.682726 | 0.736375 | 0.740868 |
| **16** | Bagging Classifier | Undersampled Data | 0.641287 | 0.801223 | 0.615746 | 0.696346 | 0.728808 |
| **17** | Decision Tree | Undersampled Data | 0.616431 | 0.765770 | 0.613396 | 0.681166 | 0.617967 |




## Final Selected Model

**Final Model:** Tuned Gradient Boosting

**Dataset Version:** Tuned Original Data

The selected model demonstrated balanced performance across precision, recall, and F1-score while maintaining strong generalization capability on unseen data.

Final selected model: Tuned Gradient Boosting

Final selected model data type: Tuned Original Data

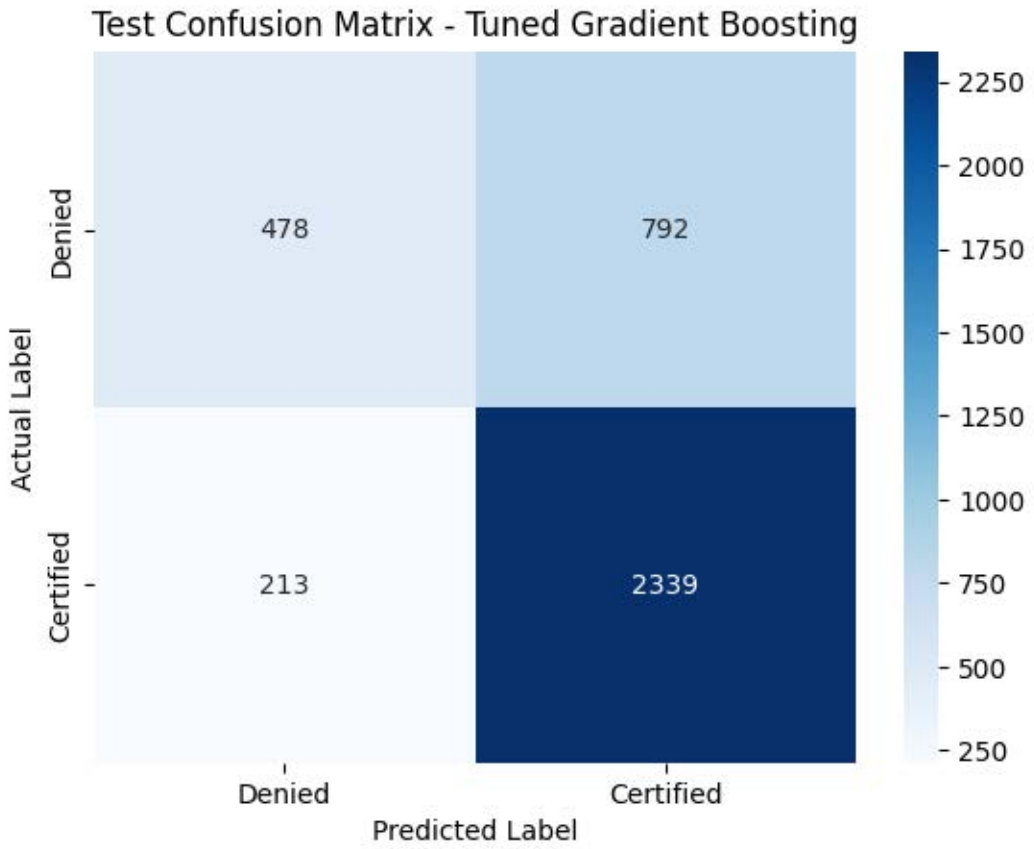
| | 18 |
|---|---|
| **Model** | Tuned Gradient Boosting |
| **Data Type** | Tuned Original Data |
| **Validation Accuracy** | 0.736525 |
| **Validation Precision** | 0.745086 |
| **Validation Recall** | 0.920486 |
| **Validation F1** | 0.82355 |
| **Validation ROC-AUC** | 0.774292 |

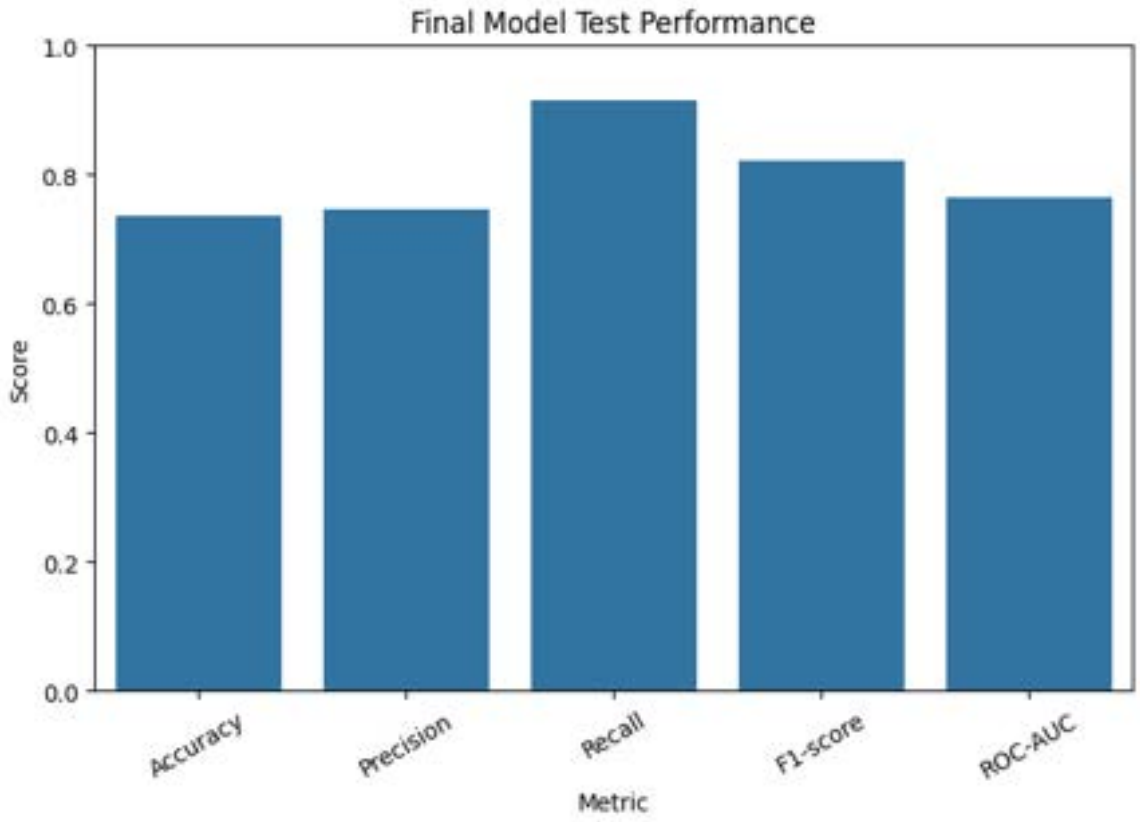
GradientBoostingClassifier

```
GradientBoostingClassifier(learning_rate=0.01, max_depth=5, min_samples_leaf=4,
                           min_samples_split=5, n_estimators=150,
                           random_state=42, subsample=0.8)
```

| | Metric | Test Score |
|---|---|---|
| **0** | Accuracy | 0.737049 |
| **1** | Precision | 0.747046 |
| **2** | Recall | 0.916536 |
| **3** | F1-score | 0.823157 |
| **4** | ROC-AUC | 0.764282 |

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| Denied | 0.69 | 0.38 | 0.49 | 1270 |
| Certified | 0.75 | 0.92 | 0.82 | 2552 |
| accuracy | | | 0.74 | 3822 |
| macro avg | 0.72 | 0.65 | 0.66 | 3822 |
| weighted avg | 0.73 | 0.74 | 0.71 | 3822 |






## 9. Actionable Insights and Recommendations

The analysis generated several business insights regarding visa certification trends.

**Key Insights**

- Applicants with higher education levels demonstrated stronger certification outcomes.
- Previous work experience positively influenced certification likelihood.
- Higher prevailing wages were associated with increased approval probability.
- Hourly wage applications displayed relatively higher denial rates.
- Certification trends varied across employment regions and continents.

**Recommendations**

- Fast-track applications with strong qualification profiles.
- Conduct deeper review for hourly wage applications.
- Validate employer information during application entry.
- Use prediction probabilities to prioritize application review queues.
- Monitor model fairness across demographic groups.
- Retrain models periodically as labor market conditions evolve.
- Maintain human oversight for final visa approval decisions.

The model can help reduce operational delays while improving consistency in visa screening workflows.

## 11. Limitations

The project has several limitations:

- The model is based on historical data and may not fully capture future immigration policy changes.
- Some variables may reflect historical biases and require fairness monitoring.
- Outliers and invalid values required preprocessing intervention.
- Model performance may decline over time if applicant trends change.
- The model should support decision-making rather than replace expert human evaluation.

## 12. Conclusion

This project demonstrates the practical application of Machine Learning in visa application screening and workforce analytics.

The analysis identified important factors influencing visa certification outcomes, including education level, work experience, prevailing wage, wage unit, and employment region.

Multiple Machine Learning models were trained and evaluated using original, oversampled, undersampled, and tuned datasets.

The final selected model demonstrated strong predictive capability and can help EasyVisa improve operational efficiency by identifying applications with high certification likelihood while flagging cases requiring additional review.

The model should be used as a decision-support tool supported by continued human oversight, fairness monitoring, and periodic retraining.

Overall, the project highlights the potential of Machine Learning to support efficient, scalable, and data-driven visa application processing.